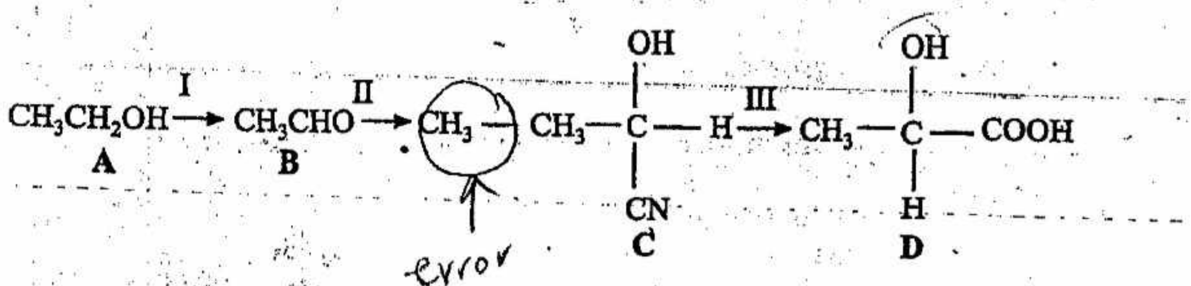
Section C

Answer at least two questions from this section.

6 Lactic acid is produced in muscles of mammals when running away from danger.

(a) Suggest reagents and conditions for each stage in the conversion of ethanol to lactic acid by the following route. [3]



(b) Which structure(s) are optically active? [1]

(c) Name the type of reaction for stage II and describe its mechanism. [4]

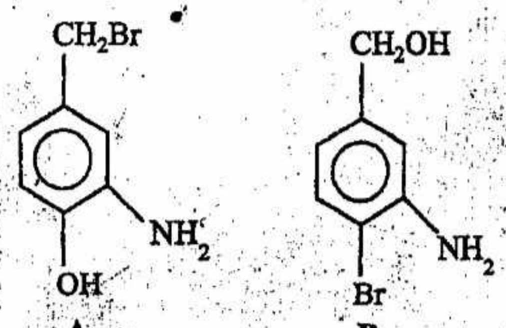
(d) (i) Name the functional groups in **D**.

(ii) Under what conditions will **D** produce a condensation polymer?

(iii) Draw **two** repeat units of the polymer and name the linkage in this polymer. [4]

[Total : 12]

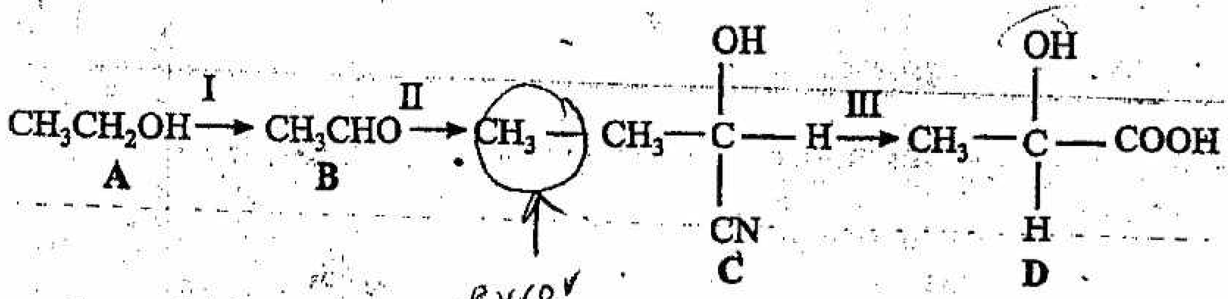
7 Compounds **A** and **B** are two isomers with different positions of functional groups.



A **B**

(a) Explain clearly, giving reagents and conditions, how the two isomers could be distinguished. [3]

(a) Suggest reagents and conditions for each stage in the conversion of ethanol to lactic acid by the following route. [3]



(b) Which structure(s) are optically active? [1]

(c) Name the type of reaction for stage II and describe its mechanism. [4]

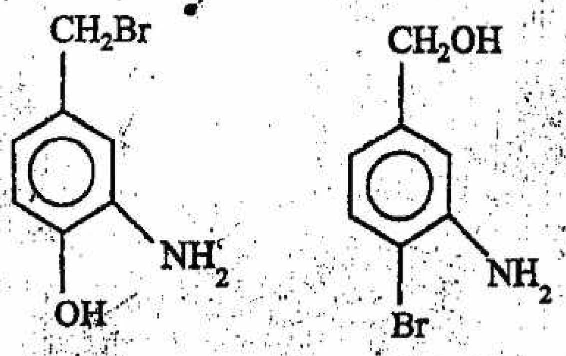
(d) (i) Name the functional groups in **D**.

(ii) Under what conditions will **D** produce a condensation polymer?

(iii) Draw **two** repeat units of the polymer and name the linkage in this polymer. [4]

[Total : 12]

7 Compounds **A** and **B** are two isomers with different positions of functional groups.



A **B**

(a) Explain clearly, giving reagents and conditions, how the two isomers could be distinguished. [3]

(b) Draw the structures obtained, if any, when both compounds react with following reagents.

(i) ethanoyl chloride

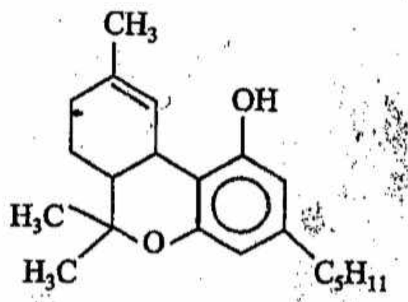
(ii) boiling aqueous sodium hydroxide

(iii) dilute nitric acid [6]

(c) Would you expect ethylamine to be a stronger or weaker base than **A**? Explain your answer. [3]

[Total : 12]

8 **(a)** *Marijuana* is a drug producing a feeling of well-being, excitement and hallucinations. Its structure is as shown below.



marijuana

(i) Name the functional groups in *marijuana.* [2]

(ii) Name the type of reaction that takes place and draw the structural formula of the organic product formed when *marijuana* reacts with following reagents.

1 aqueous bromine (in the dark)

2 cold aqueous sodium hydroxide

3 cold dilute $KMnO_4$ [8]

(b) Describe a simple test tube reaction you could carry out to distinguish between the two ketones **A** and **B** below.

$CH_3CH_2C(=O)CH_2CH_3$ **A**

$CH_3C(=O)CH_2CH_2CH_3$ **B**

[2]

[Total : 12]

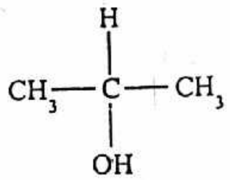
REM

Section C

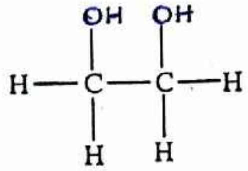
Answer at least two questions from this section.

6 The structures and boiling points of three alcohols are given below.

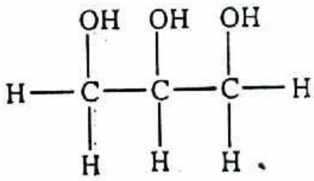
Alcohol — Boiling Point

I  83°C

propan -2-ol

II  198°C

ethane 1,2-diol

III  290°C

propan 1,2,3 triol

(a) How can the differences in boiling points be explained? [2]

(b) **(i)** What is observed when each of these compounds reacts with I_2 in aqueous NaOH?

(ii) Give the structural formulae of the organic product(s) formed from propan-2-ol. [4]

(c) Ethane -1, 2- diol, is one of the two monomers used to make terylene.

(i) Give the structure of the other monomer.

(ii) Draw the repeat unit for terylene and name the linkage formed.

(iii) What type of polymer is terylene? [4]

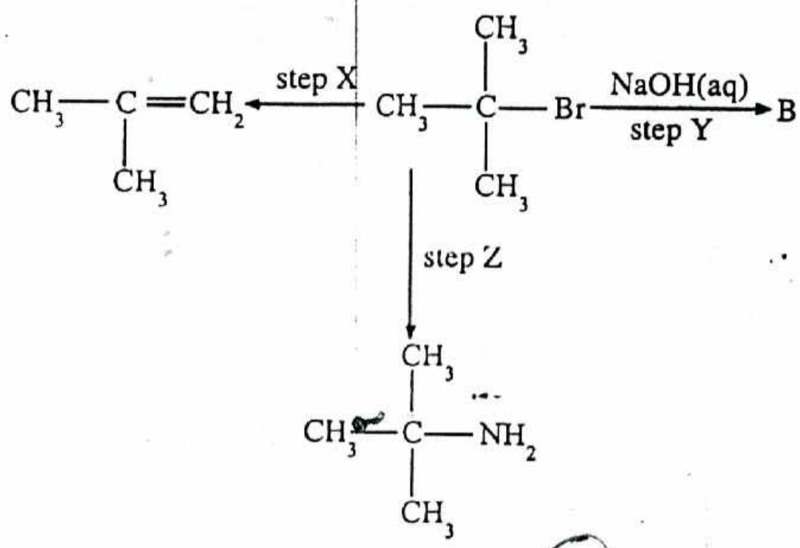
(d) Give the structural formula and name the organic product formed when propan-2-ol is warmed with acidified potassium dichromate (VI). [2]

[Total: 12]

Halogens react with organic compounds to give halogenoalkanes.

(a) State and explain the order of the relative C–X bond strengths of the halogenoalkanes, where X is the halogen. [2]

(b) Study the reaction scheme shown and answer the questions below.

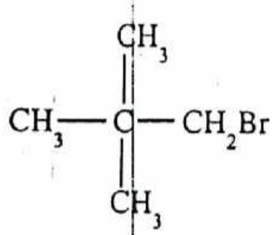


(i) Give the structural formula of B. 1

(ii) State the reagents and conditions used in steps X and Z. 2

(iii) Name the types of reactions in steps X and Y. 2

(iv) Describe the mechanism involved when the compound below reacts with NaOH(aq):



3

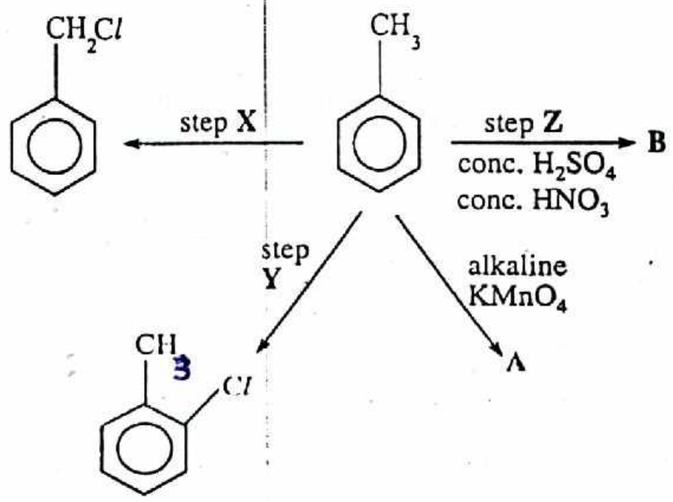
[8]

(c) Name one halogenoalkane and its harmful effect. [2]

[Total: 12]

8 (a) Explain why methylbenzene is more reactive than benzene. [1]

(b) Study the reaction scheme shown, and answer the questions that follow.



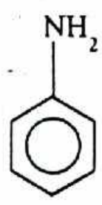
(i) Give the structural formulae of compounds A and B. 2

(ii) State the reagents and conditions for steps X and Y. 2 [4]

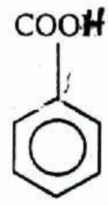
(c) What types of reactions are occurring at steps X, Y and Z? [3]

(d) What is the effect of the methyl group on the position of substitution? [1]

(e) Giving an explanation for your answer, compare the rate of nitration of the following compounds relative to benzene.



phenylamine



benzoic acid

Give the structural formula of the mononitrated product of benzoic acid. [3]

[Total: 12]

(c) ClO_3^- and ClO_4^- are some of the oxyanions that can be formed by chlorine. Suggest their shapes and state the bond angles. [4]

(d) When $NaClO_3$ is heated to its melting point it disproportionates.

(i) What is disproportionation?

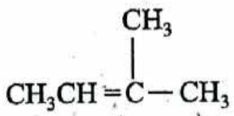
(ii) Suggest an equation for the disproportion that occurs. [2]

[Total: 12]

Section C

*Answer at least **two** questions from this section.*

6 A certain alkene is believed to have either structure **A** or **B** below.



A

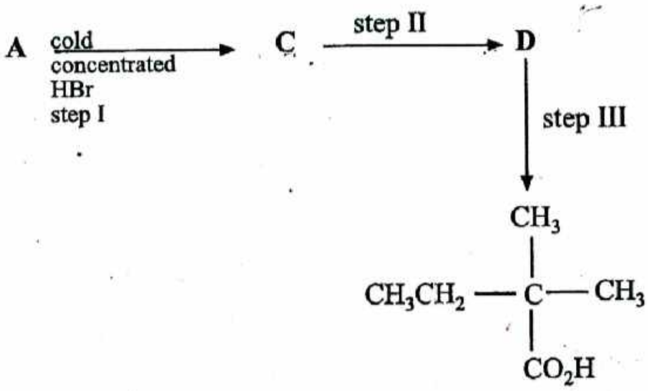
$CH_3-CH=CHCH_2CH_3$

B

(a) Describe simple test tube reactions that can be carried out to verify the structure of the alkene.

Limit the answer to a two step procedure. [3]

(b) Alkene **A** can undergo the changes shown in the scheme below:



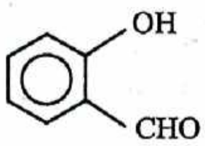
(i) Suggest reagents and conditions for steps II and III. [4]

(ii) Draw structural formulae for the intermediates **C** and **D**. [2]

(iii) State **one** physical property in which **A** and **B** would differ and explain the difference. [3]

[Total: 12]

7 Compounds **E**, **F** and **G** are isomers of each other. Their structures are shown below.



E

F

G

(a) **(i)** What types of structural isomerism do they exhibit?

(ii) Arrange these compounds in order of increasing boiling point and explain your order. [5]

(b) Give the structure of the organic product and state what would be observed when compound F reacts with the reagents listed below.

(i) $Br_{2(aq)}$

(ii) Tollen's reagent

(iii) 2,4-dinitrophenyl hydrazine.

[6]

(c) Name the type of reaction that occurs in **(b)(iii)**. [1]

[Total: 12]

8 Suggest explanations for the following observations:

(a) Carbonyl compounds react with HCN in the presence of bases but not acids. [2]

(b) The reaction of ethene with bromine dissolved in aqueous sodium chloride produces 1-bromo-2-chloro ethane as one of the products. [2]

(c) The pK_a of $CH_2(F)COOH$ is 2.66 and that of $CH_2(I)COOH$ is 3.12. [2]

(d) When I-hydroxy propenol, $CH(OH)CHCH_3$, is reacted with HBr the resultant mixture is optically inactive. [2]

(e) $CH_3CO\overset{NH_2}{\cancel{HN_2}}$ is neutral whereas $CH_3CH_2\overset{N}{\cancel{B}}H_2$ is basic. [2]

(f) CH_3COCl is more acidic in aqueous solution than CH_3CO_2H. [2]

[Total: 12]

SECTION C

Answer at least two questions from this section.

6 (a) Explain the relative acidities of water, phenol and ethanol. [3]

(b) In the first "breathalyser" test for alcohol, drivers were asked to expel air from their lungs into acidified potassium dichromate crystals.

State and explain what would be observed when the test was positive. [3]

(c) 2, 4, 6 - trichlorophenol which is present in some antiseptics such as Dettol is an example of a substituted phenol.

(i) Draw the displayed structural formulae of **two** isomers of 2, 4, 6 - trichlorophenol.

(ii) State reagent(s) and condition(s) necessary to prepare 2, 4, 6-trichlorophenol from phenol. [3]

(d) Propane-1, 2, 3- triol is usually known by its common name glycerol.

Describe the reactions of propane -1, 2, 3-triol with

(i) sodium,

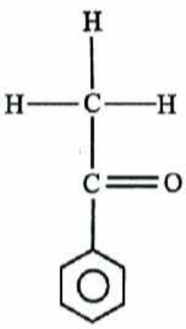
(ii) ethanoyl chloride.

Give the structure of the organic product formed with each reagent. [3]

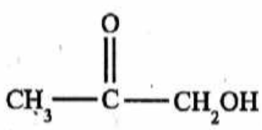
[Total: 12]

7 (a) Name the following organic compounds:

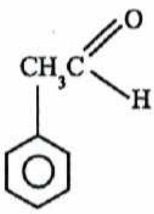
A



B



C



[3]

[Turn over

(b) Give the structural formulae of organic products formed when ***B*** reacts with the following reagents:

(i) 2,4 – dinitrophenyl hydrazine

(ii) $K_2Cr_2O_7$ in dilute H_2SO_4

(iii) HCN

(iv) $I_{2(aq)}$ and $NaOH_{(aq)}$

What conditions are necessary for the reaction in (i)? [5]

(c) Aldehydes, $RCHO$, undergo both reduction and oxidation. The processes being represented by the following half equations:

Oxidation: $RCHO + H_2O \rightarrow RCOOH + 2e^- + 2H^+$

Reduction: $RCHO + 2H^+ + 2e^- \rightarrow RCH_2OH$

The oxidation half equation of the reducing agent, $LiAlH_4$, is shown below.

$4H_2O + LiAlH_4 \rightarrow LiOH + Al(OH)_3 + 8e^- + 8H^+$

(i) Deduce a balanced equation for the reduction of an aldehyde, *R-CHO*, by $LiAlH_4$.

(ii) Aldehydes such as $HCHO$ and C_6H_5CHO can, under certain conditions oxidise and reduce each other. Given that oxidation involves the nucleophilic attack of the carbonyl carbon suggest with an explanation, which of the two aldehydes is oxidised and which is reduced.

Write an equation for the reaction. [4]

[Total: 12]

8 **(a)** The cyanide ion, CN^-, reacts with both chloropropane and propanone.

(i) State reagents and conditions for each reaction.

(ii) Name the type of each reaction.

(iii) Name each organic product.

(iv) Give a reaction mechanism for each reaction. [10]

(b) Chlorobenzene unlike chloropropane does not react with the cynide, CN^-, ion under similar conditions.

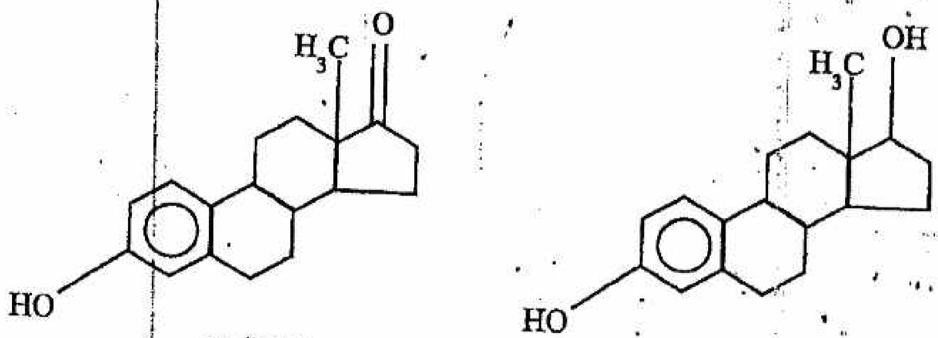
Suggest **two** possible explanations for this observation. [2]

[Total: 12]

Section C

Answer at least two questions from this section

6 Fig.1 shows the structure of two steroids, oestrone and oestradiol.



oestrone oestradiol

Fig. 1

(a) Suggest how you would distinguish between the two structures using their

(i) physical properties,

(ii) chemical properties.

[4]

(b) Describe how oestrone can be converted into oestradiol? [2]

(c) The two hydroxyl groups in oestradiol may under-go similar and different reactions.

Describe a reaction, giving reagents, conditions, observations and drawing the structure of the organic product formed when they react

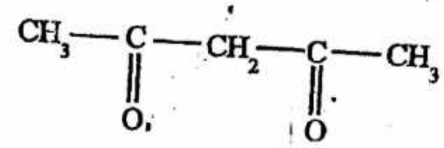
(i) similarly,

(ii) differently.

[6]

[Total: 12]

7. Fig. 2.1 shows the structure of pentan -2, 4 - dione. The compound exhibits a phenomenon called tautomerism in which keto-enol equilibrium is established. The enol form contains a carbon to carbon double bond.



pentan – 2, 4 – dione

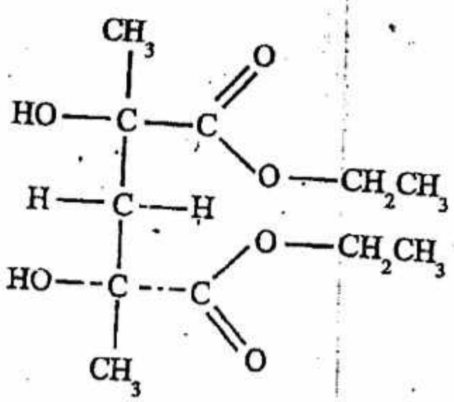
enol → -ene + -ol

Fig. 2.1

(a) Suggest a structural formula for the enol form. [1]

(b) Describe a simple test-tube reaction that can be used to distinguish between the two forms giving the structure of the organic products formed. [3]

(c) Fig. 2.2 shows the structure of compound **A**.



compound **A**

Fig. 2.2

Design a 3-stage synthetic route for compound **A** using pentane - 2, 4 - dione as the starting material.

For each step:

(i) give reagents and conditions

(ii) name the type of reaction

(iii) draw the structural formula of the intermediate product [8]

[Total: 12]

8 **Fig. 3** shows the structure of a compound, secreted by animals, that influences the behaviour of other members of the same species, and may be involved in sexual attraction and territorial marking.

$$CH_3COCH_2CH_2CH_2CH_2CHCHCO_2H$$

Fig. 3

(a) (i) Identify the functional groups present in this compound.

(ii) Suggest test-tube reactions that can be used as confirmatory tests for the presence of each functional group. [8]

(b) A 0.552 g sample of the compound was treated under reflux with hot, concentrated $KMnO_4$ solution. The products were completely reacted with 4.50 cm^3 of 2.0 $mol dm^{-3}$ sodium hydroxide solution.

Deduce the oxidation products of the compound. [4]

[Total: 12]

Section C

*Answer at least **two** questions from this section.*

6 **(a)** Methylpropene, $(CH_3)_2CCH_2$, reacts with hydrogen bromide.

(i) Name, with a reason, the type of reaction undergone by methylpropene and hydrogen bromide.

(ii) Describe the mechanism for the reaction between methylpropene and hydrogen bromide.

(iii) Draw the displayed structural formula of the product.

(iv) Name the product of this reaction.

(v) Use your answer to **(a)(ii)** to construct a labelled reaction pathway diagram for the reaction.

[9]

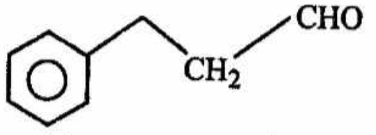
(b) The halogenoalkane 2–chlorobutane reacts with ethanolic aqueous sodium hydroxide to produce a mixture of three alkenes with the same molecular formula as methylpropene.

(i) State the type reaction between 2–chlorobutane and ethanolic $NaOH_{(aq)}$.

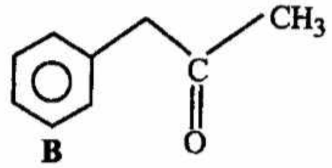
(ii) Draw the displayed structural formula of any **two** alkenes produced in this reaction.

[3]
[Total: 12]

7 **(a)** Compounds **A** and **B** are structural isomers:



A



B

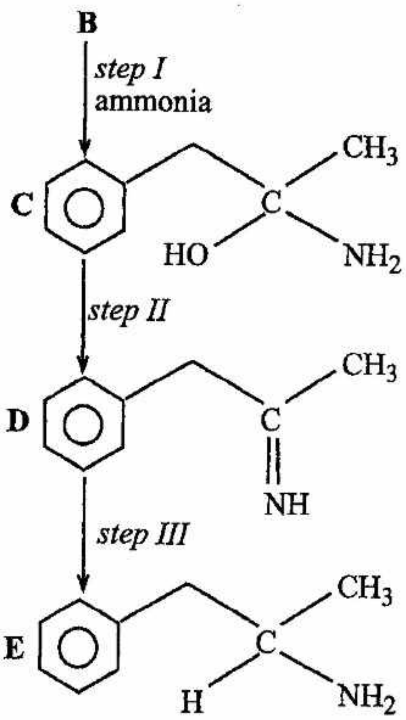
(i) Write the molecular formula of **A**.

(ii) Name the type of isomerism shown by the two compounds.

(iii) Describe a simple test tube reaction that can be used to distinguish **A** from **B**.

[5]

(b) **B** can react with ammonia the same way it does with hydrogen cyanide. The following steps show the conversation of **B** into **E**.



(i) Suggest the type of reaction in

1. *step I*

2. *step II*

(ii) Give the reagents and conditions for step *III*.

[3]

(c) Compound **C** can react with ethanoyl chloride in a ratio of 2: 1.

(i) State the conditions used in this reaction.

(ii) Describe the observations made.

(iii) Draw the displayed structural formula of the organic product.

(iv) Name **one** functional group present in the product.

[4]
[Total: 12]

8 **(a)** Define the terms

(i) *polymer,*

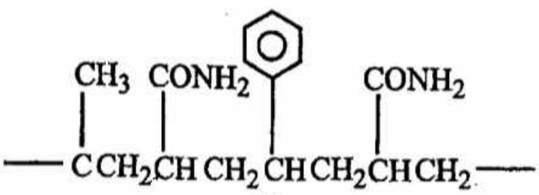
(ii) *monomer* and

(iii) *polymerisation.* [3]

(b) **X** and **Y** are segments of synthetic polymers.

~$CONHCH_2CH_2NHCOCONH$~

X



Y

(i) Predict the type of polymerisation which produces polymer

1. **X,**

2. **Y.**

(ii) Name the type of linkage in **X.**

(iii) Deduce the structures of monomers in

1. **X,**

2. **Y.**

[7]

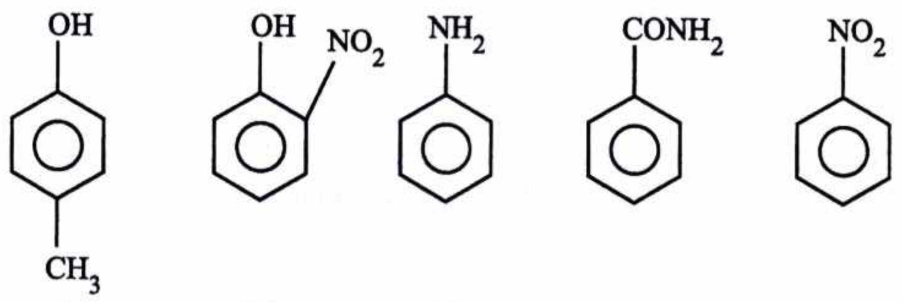
(c) State **two** difficulties associated with the disposal of polymers. [2]

[Total: 12]

Section C

*Answer at least **two** questions from this section.*

6 **Fig. 1** shows organic compounds **A, B, C, D** and **E**.



A **B** **C** **D** **E**

Fig. 1

(a) State, with a reason, the compound that is

(i) most acidic,

(ii) most basic,

(iii) neutral.

[3]

(b) Compare the reactions of **C** and **E** with bromine by

(i) stating conditions for each reaction,

(ii) giving the structural formula of the organic product.

[4]

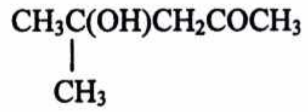
(c) Two of the compounds react to form a dye in a two step reaction.

(i) Write the **two step** reaction scheme to show formation of the dye.

(ii) Give reagents and conditions for step one.

[5]
[Total: 12]

7 The structural formula of an organic compound **A** is shown below



(a) (i) Name compound **A**.

(ii) Draw the skeletal formula of **A**.

[2]

(b) (i) Name the **two** types of isomers in which the isomers of **A** have the same structural formula but differ in orientation of atoms in space.

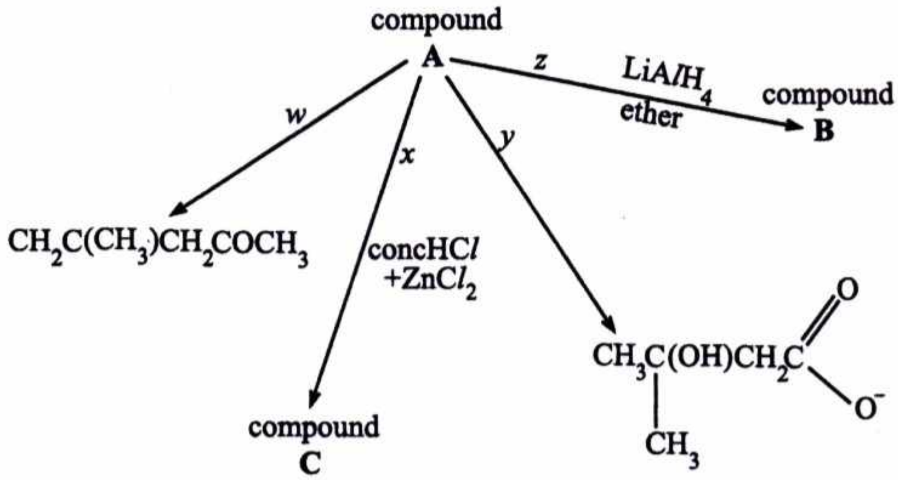
(ii) Suggest structural formula of isomers of **A**, with five carbon atoms in the parent chain, that would exhibit

1. optical isomerism,

2. geometric isomerism.

[4]

(c) Compound **A** undergoes the reactions shown.



(i) Give reaction conditions for steps

1. *w*,

2. *y*.

(ii) Draw the structural formula of

1. **B**,

2. **C**.

(iii) Describe the observation made when reaction x occurs.

(iv) Suggest an explanation for the observation in **(c)(iii)**.

[6]
[Total: 12]

8 **(a)** Explain

(i) the solubility of amines in water,

(ii) why higher members of amines are less soluble in water,

(iii) why higher members of amines are less odorous.

[3]

(b) The table shows pK_b values of phenylamine and methylamaine.

	pK_b value
phenylamine	9.30
methylamine	3.36

(i) With the aid of an equation explain the basicity of amines.

(ii) Explain the differences in the pK_b values of phenylamine and methylamine.

[5]

(c) A basic nitrogen compound of the formula $C_8H_{11}N$ reacts with benzene diazonium chloride to give an orange precipitate. It does not react with potassium manganate (VII) but decolourises bromine water giving a white precipitate.

(i) Suggest a structural formula for $C_8H_{11}N$.

(ii) State the conditions used in the reaction with benzene diazonium chloride.

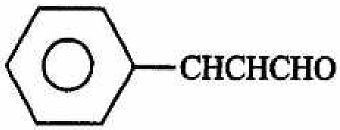
(iii) Write the equations for the reactions described, giving full structural formulae of products formed.

[4]
[Total: 12]

Section C

Answer at least two questions from this section

6 (a) **Fig. 6.1** shows the structure of cinnamaldehyde.



cinnamaldehyde

Fig. 6.1

(i) Name

1. any **two** functional groups present in cinnamaldehyde,
2. the type of isomerism that cinnamaldehyde exhibits.

(ii) Draw structures to show the isomers of cinnamaldehyde.

[4]

(b) (i) Describe the reaction mechanism of cinnamaldehyde with HCN.

(ii) Explain why the reaction in (b)(i) is carried out in the presence of aqueous NaOH.

(iii) Give the structure of the organic product obtained when the compound produced in **(b)(i)** undergoes hydrolysis in aqueous NaOH. [6]

(c) Describe a test giving the reagents, conditions and observations, to distinguish between benzaldehyde and cinnamaldehyde. [2]

[Total:12]

7 (a) **Fig. 7.1** shows how benzene can be converted into other organic aromatic compounds.

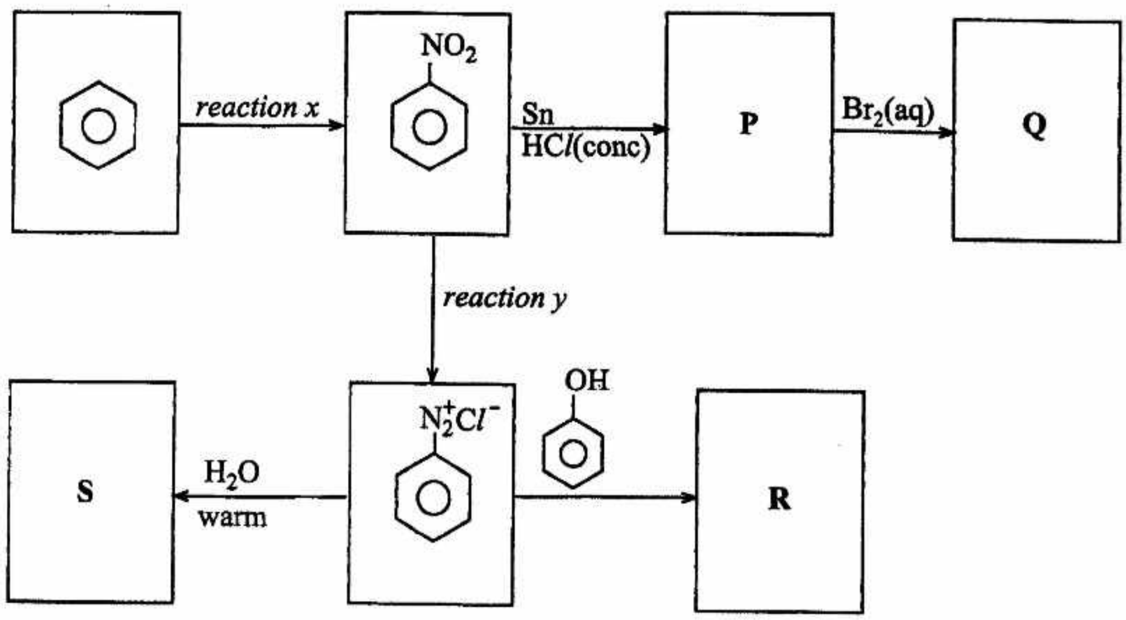


Fig. 7.1

(i) Give reagents and conditions for

1. *reaction x,*

2. *reaction y.*

(ii) Draw the displayed structural formulae of compounds **P, Q, R** and **S**.

(iii) Name the type of reaction that forms **R.**

(iv) State **one** use of **R**.

[8]

(b) State and explain what happens when

1. benzene diazonium chloride is placed in warm water,

2. ethylamine, dissolved in hydrochloric acid, is evaporated. [4]

[Total:12]

8 (a) **Fig. 8.1** shows the formula of a synthetic polymer **X**.

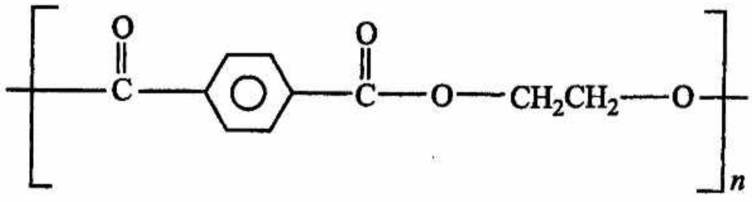


Fig. 8.1

(i) Draw and name the structures of the monomers that react to form **X**.

(ii) Name the type of polymerisation that results in formation of **X**.

(iii) With the aid of a chemical equation explain why a fabric made of **X** is weakened when placed in aqueous NaOH.

[7]

(b) Describe and explain what happens when

(i) a mixture of ethanol and acidified sodium dichromate (VI) solution is heated,

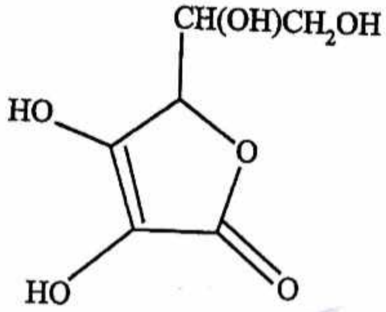
(ii) excess chlorine is added to boiling methylbenzene in the presence of UV light.

[5]
[Total:12]

Section C

Answer at least two questions from this section

6 (a) **Fig. 6.1** shows the structure of ascorbic acid, a white crystalline compound found dissolved in the juices of fresh fruits and vegetables.



ascorbic acid

Fig. 6.1

(i) Name

1. any **two** functional groups in ascorbic acid,

2. the type of isomerism exhibited by ascorbic acid.

(ii) Draw the structures of isomers of ascorbic acid.

(iii) Explain why ascorbic acid is soluble in water. [6]

(b) (i) State the observations made, when ascorbic acid is reacted with

1. cold dilute $KMnO_4$,

2. hot concentrated $KMnO_4$,

3. phosphorus pentachloride.

(ii) Give the displayed structural formulae of the organic product formed for each of the reactions in **(i)**.

[6]
[Total:12]

7 (a) The molecular formula of a compound, **A**, is C_6H_7N. When **A** is reacted with excess hydrochloric acid and sodium nitrite below 5 °C, an organic compound, **B**, is formed which produces phenol when poured into hot water.

(i) Deduce the structural formula of

1. **A**,

2. **B**.

(ii) Write the chemical equation for the reaction of **B** with hot water.

(iii) Give a **two-step** reaction for the formation of **A** from benzene. State the reagents and conditions for each step.

[6]

(b) **Fig. 7.1** shows how phenol can be converted to other organic compounds, **E** and **F**.

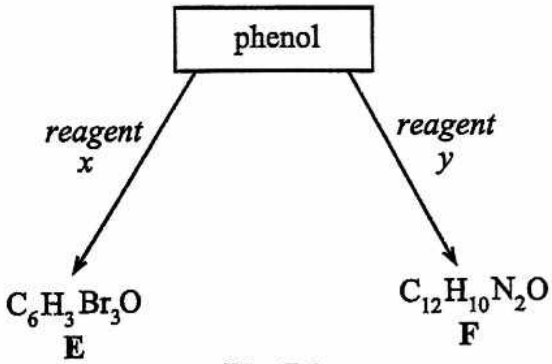


Fig. 7.1

(i) Give the structures of

1. **E**,

2. **F**.

(ii) Name the type of reaction undergone by phenol with *reagent y* and state the conditions used.

(iii) Describe the observations made during the production of **E**.

[6]
[Total:12]

8 (a) State any **two** differences between addition and condensation polymerisation. [2]

(b) (i) Draw **two** repeat units of

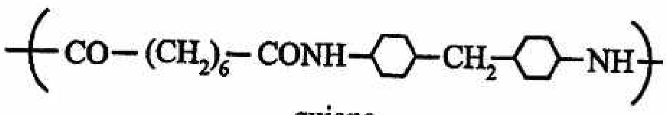
1. nylon 6,

2. nylon 66.

(ii) Deduce from the units in **(i)** **one** difference between the two polymers.

[5]

(c) Quiana is a polymer that has a silk-like feel while polyacrylonitrile is a fabric which is stronger than wool. The structure of quiana is shown



quiana

(i) Name the type of linkage that is present in quiana.

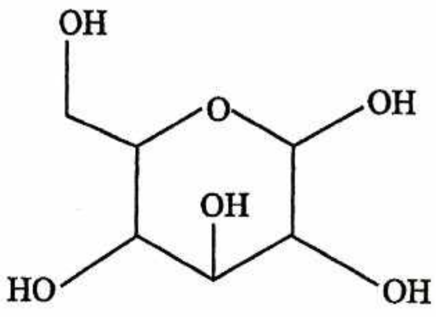
(ii) Draw **two** repeat units of polyacrylonitrile given that the structure of acrylonitrite is $CH_2 = CH-C \equiv N$,

(iii) Compare the reactivities of quiana and polyacrylonitrile with aqueous sodium hydroxide.

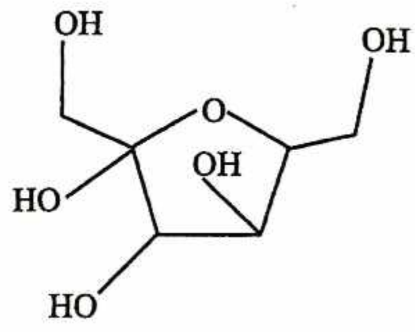
(iv) Draw the structure(s) of any organic product(s) formed in **(iii)**.

[5]
[Total: 12]

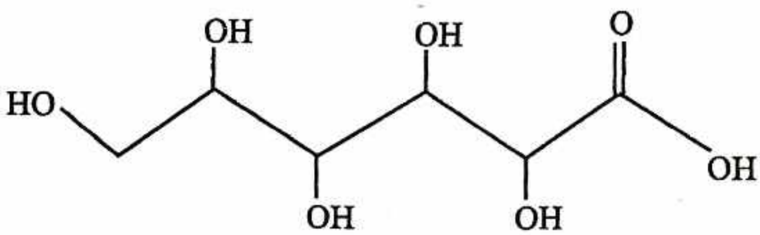
7 **(a)** **Fig. 7.1** shows the structures of glucose, fructose and gluconic acid, the major constituents of honey. Bees make honey by concentrating the sugars and removing excess water in nectar.



glucose



fructose



gluconic acid

Fig. 7.1

(i) State the molecular formula of

1. glucose,

2. fructose.

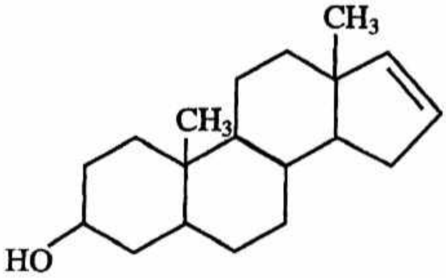
(ii) Deduce, from **(i)** the relationship between glucose and fructose.

(iii) State, with reasons, the effect of honey on plane polarised light.

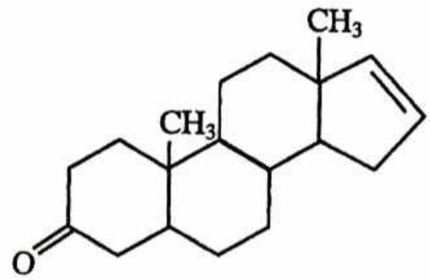
(iv) Suggest why honey, when kept at normal temperature, does not get bad over a long period of time.

[6]

(b) **Fig. 7.2** shows the structural formula of androstenol and androstenone.



androstenol



androstenone

Fig. 7.2

(i) State the number of chiral centres in androstenone.

(ii) Describe any **one** chemical test that can be used to distinguish between the two compounds.

(iii) Suggest how androstenone can be converted to androstenol.

(iv) Describe the mechanism for the reaction of aqueous bromine with androstenone.

[6]
[Total: 12]

8 (a) The reaction scheme, in **Fig. 8.1** shows the synthesis of 4-nitrophenylamine from phenylamine.

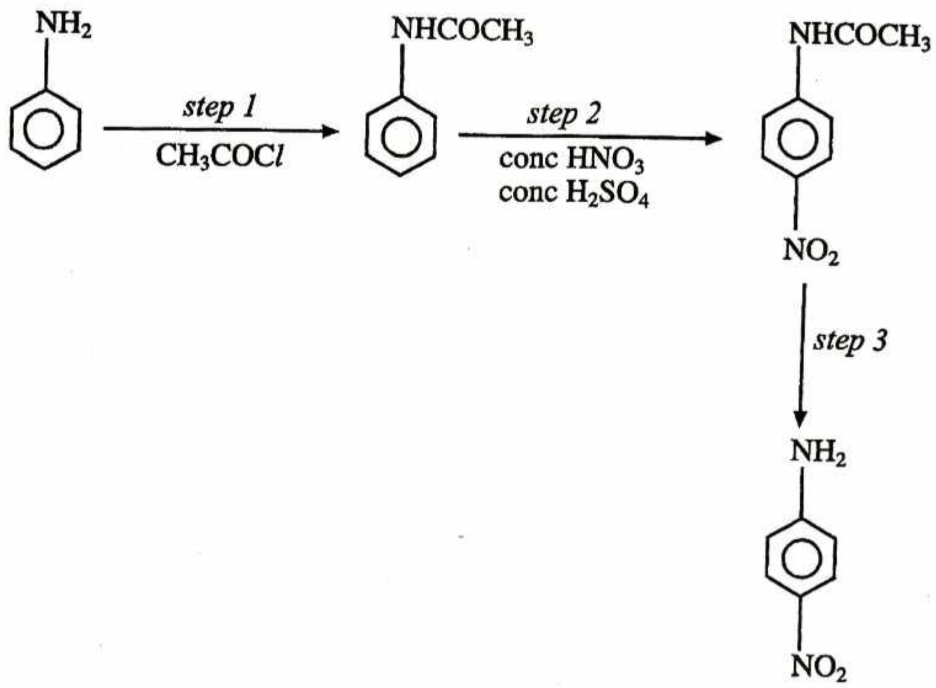


Fig. 8.1

(i) Name the reaction occurring in

1. *step 1*,

2. *step 2*.

(ii) State reagents and conditions for *step 3*.

(iii) Outline the mechanism for the reaction in *step 2*.

[6]

(b) Outline a three step synthesis of the compound, $CH_3CH_2NH_2$ using methane as the starting material.

[5]

(c) Explain why ethylamine is a base.

[1]

[Total: 12]

Section C

Answer at least two questions from this section.

6 **(a)** Organic compounds, **P**, **Q** and **R** are non-cyclic branched hydrocarbons of the formula C_6H_{12}. **P** shows optical isomerism while **Q** and **R** show geometrical isomerism.

Deduce the possible structures of **P**, **Q** and **R**. [3]

(b) **Fig. 6.1** shows the structure of an organic compound **Z**.

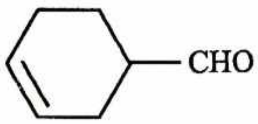


Fig. 6.1

(i) Draw the structures of the organic products obtained when

1. a few drops of Fehlings solution are added to **Z**,

2. **Z** reacts with cold alkaline $KMnO_4$.

(ii) State the observable changes that occur in each of the reactions in **(i)**.

(iii) Describe, with the aid of chemical equations, the changes that occur when hydrogen cyanide is added to **Z** under acidic conditions.

[6]

(c) **Fig. 6.2** shows the structure of a triester.

$$\begin{array}{r} CH_3-(CH_2)_6-COOCH_2 \\ | \quad \\ CH_3-(CH_2)_6-COOCH \;\; \\ | \quad \\ CH_3-(CH_2)_6-COOCH_2 \end{array}$$

Fig. 6.2

(i) Draw the structures of the organic products formed when the triester is heated under reflux with an excess of aqueous sodium hydroxide.

(ii) Compare the solubilities, in water, of the organic products in **(i)** and the triester.

[3]

[Total: 12]

Section C

Answer at least two questions from this section.

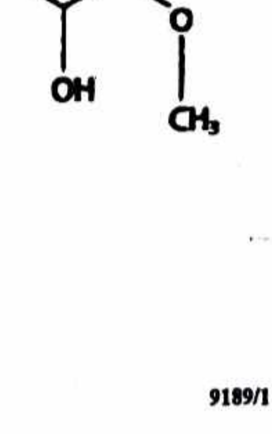
6. **(a) Define the term functional group** **[1]**

(b) Describe the presence of each of the following functional groups affect the reactivity of benzene

(i) - NH_2

(ii) - CHO **[4]**

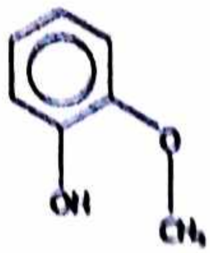
(c) Oil of cloves is a traditional remedy for toothache; it is a topical treatment, which means that it is applied externally. The main active ingredient of oil of cloves is eugenol, which also gives cloves their characteristic smell. The structure of eugenol is shown below



eugenol

8

A student suggested that eugenol could be prepared by an electrophilic substitution of 2-methoxyphenol, which is produced in the gut of desert locusts. This compound is one of the main components of the pheromones that cause locust swarming. The structure of 2-methoxyphenol is shown below.



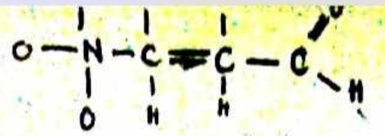
2-methoxyphenol

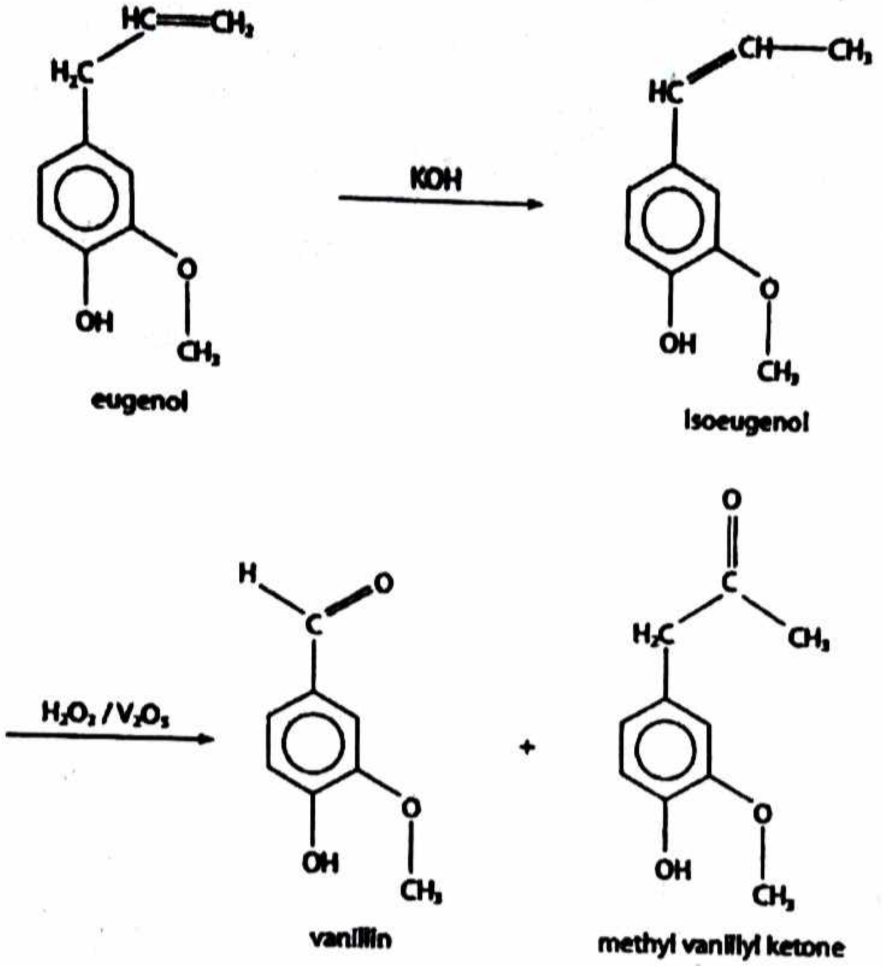
(i) Use your knowledge of electrophilic substitution reactions to suggest the structure of an electrophile that might be used in this reaction [1]

(ii) Suggest an organic compound that could be used to produce the electrophile that you have given in (a)(i). [1]

(iii) Write the mechanism for the electrophilic substitution to prepare eugenol from 2-methoxyphenol, using the electrophile that you have given in (a)(i). [3]

(d) Eugenol has been used in the manufacture of vanillin, the compound responsible for the flavour of vanilla. The process involves the conversion of eugenol to its isomer isoeugenol, which is then oxidised by hydrogen peroxide with a vanadium(V) oxide catalyst.





Isoeugenol exists as two stereoisomers, whereas eugenol has just one structure. State the type of stereoisomerism shown by isoeugenol and explain why it can show this type of stereoisomerism, whereas eugenol does not. [2]

[Total:12]

7. An organic compound A has a molecular formulae $C_{10}H_{18}$. It readily decolourises bromine reacting in the ratio 1:2.When A is treated with hot concentrated potassium permanganate (VII) three compounds are produced.

B $C_3H_4O_2$ C, $C_5H_8O_2$ D, C_3H_6O

B reacts with sodium carbonate whereas C and D do not.

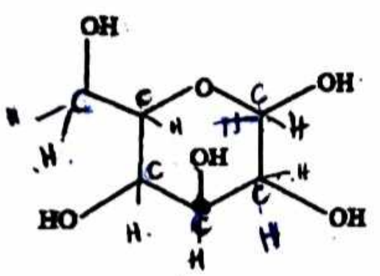
C and D both react with 2.4-dinitrophenyl hydrazine but not with Fehlings solution.

(a) Give the structural formulae of A to D and write equations for all the reactions involved.[8]

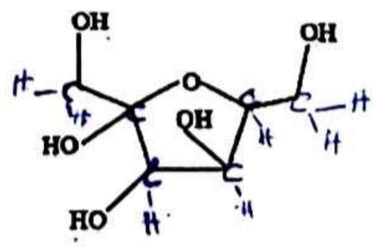
(b) D also reacts with hydrogen cyanide in the presence of traces sodium cyanide. Name the type of reaction undergone and give its mechanism. [4]

[Total:12]

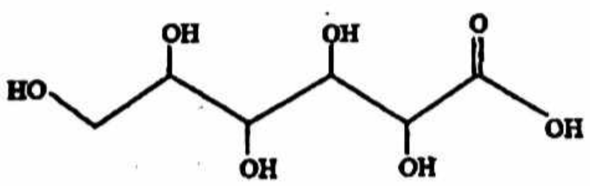
8. (a) The diagram below shows the structure of glucose fructose and gluconic acid, the major constituents of honey. Bees make honey by concentrating the sugars and removing excess water in nectar



glucose



fructose



gluconic acid

(i) State the molecular formula of

1. Glucose
2. Fructose

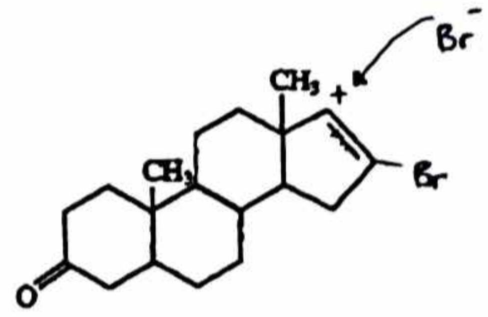
(ii) Deduce from (i) the relationship between glucose and fructose

(iii)State with reasons the effect of honey on plane polarised light

(iv)Suggest why honey when kept at normal temperature, does not get bad over a long period of time. [6]

(b) The diagram below shows the structural formula of androstenol and androstenone

androstenol



androstenone

(i) State the number of chiral centres in androstenone

(ii) Describe any one chemical test that can be used to distinguish between the two compounds.

(iii) Suggest how androstenone can be converted to androstenol

(iv) Describe the mechanism for the reaction of aqueous bromine with androstenone [6]

[Total:12]

End of Examination

Section C

*Answer at least **two** questions from this section.*

6 A carboxylic acid, **P**, has the following composition by mass

carbon	-	62.10%
oxygen	-	27.60%
hydrogen	-	10.30%

Neutralisation of a sample of 0.10 g of **P** requires 8.60 cm^3 of 0.10 $moldm^{-3}$ sodium hydroxide.

(a) Calculate the

(i) empirical formula of **P**,

(ii) molecular formula of **P**.

[5]

(b) **P** can be synthesized in the laboratory from 1-bromopentane by a two stage reaction sequence.

$$CH_3(CH_2)_4Br \xrightarrow{\text{stage 1}} \mathbf{Q} \xrightarrow{\text{stage 2}} \mathbf{P}$$

(i) Suggest the reagents and conditions for each stage.

(ii) Name the reaction that occurs in *stage 2*.

(iii) Deduce the structural formula of

1. **P**,

2. **Q**.

[7]
[Total: 12]

7 **Fig. 7.1** shows the structure of lanosterol.

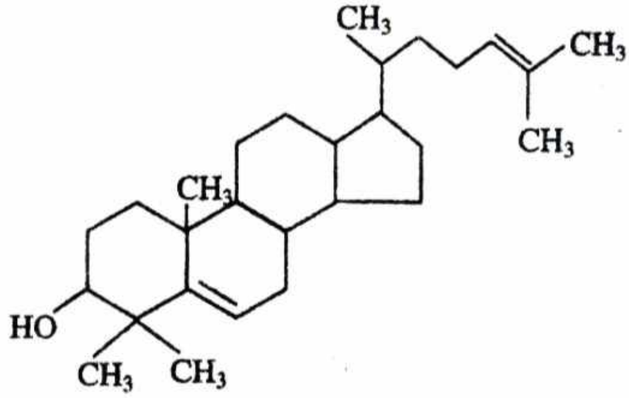


Fig. 7.1

(a) State the number of chiral centres present in this molecule. [1]

(b) **(i)** Name any **two** functional groups present in lanosterol.

(ii) Describe reactions that can be carried out to confirm the the presence of the functional groups named in **(b)(i)**.

[6]

(c) Give the structural formula(e) of the organic product(s)formed when lanosterol reacts with

(i) hot potassium permanganate under reflux,

(ii) phosphorus pentachloride,

(iii) cold potassium permanganate.

[5]
[Total: 12]

8 The structures of aspirin and paracetamol are shown in **Fig. 8.1**.

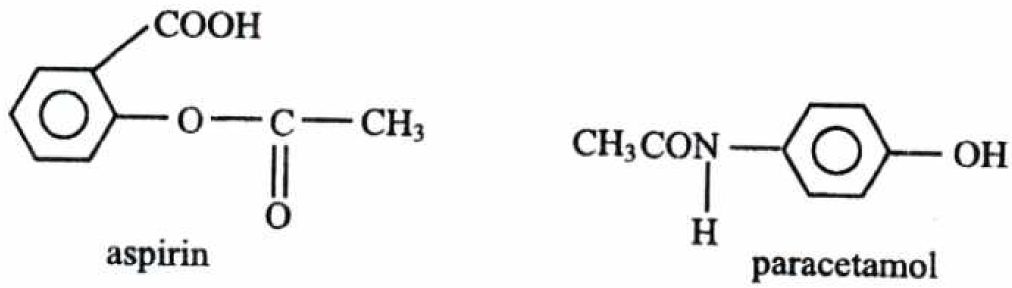


Fig. 8.1

(a) Name the linkage in

1. paracetamol,

2. asprin.

[2]

(b) **(i)** Draw the structural formulae of organic products formed when compounds in **Fig. 8.1** are hydrolysed.

(ii) State the reagent and conditions used in the hydrolysis reaction.

[5]

(c) Describe a reaction to distinguish paracetamol from aspirin. [2]

(d) Comment on the solubility of paracetamol in water. [2]

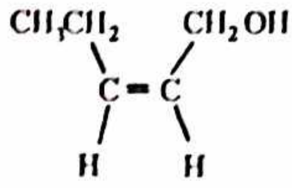
(e) Draw the structural formula of the compound formed when paracetamol reacts with aspirin. [1]

[Total: 12]

Section C

Answer any two questions from this section.

Fig.6.1 shows the structure of an organic compound **C**.



compound **C**

Fig.6.1

(a) Draw the structure of the organic product formed when **C** reacts with

(i) a solution of bromine in tetrachloromethane,

(ii) CH_3COOH in the presence of an acid catalyst.

[2]

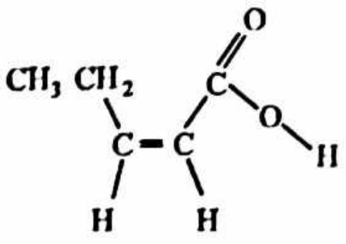
(b) Compounds **D** and **E** were formed when **C** was reacted with HBr.

(i) Draw structures of **D** and **E**.

(ii) Explain how **D** and **E** are related.

[3]

(c) Fig.6.2 shows the structure of a compound **F** that can be prepared from compound **C**.



F

Fig.6.2

(i) Suggest reagents for the conversion of **C** to **F**.

(ii) Identify the intermediate for the reaction in **c (i)**.

(iii) Describe how the intermediate can be removed from the reaction mixture

(iv) State the type of reaction that produces the intermediate.

[4]

(d) (i) Describe the bonding in benzene in terms of sigma and pi bonds.

(ii) Outline the mechanism for the reaction of benzene with $Cl_{2(g)}$ in the presence of $FeCl_3$ catalyst.

[6]

[Total:15]

7 Fig.7.1 shows a reaction pathway of a polymer.

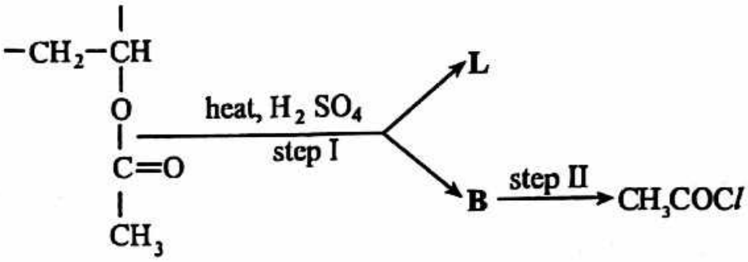


Fig.7.1

(a) (i) Draw the structural formula of

1. the monomer from which the polymer was made,

2. **L**,

3. **B**.

(ii) Identify any **two** functional groups in the monomer.

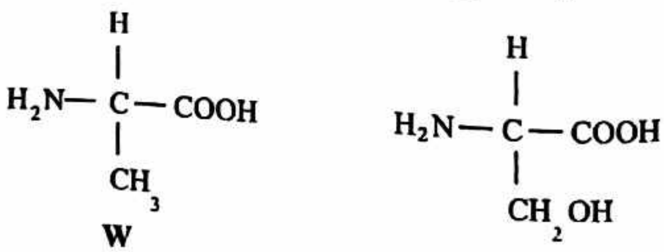
(iii) The polymer was reacted with excess sodium hydroxide to form another polymer.

Draw the repeat unit of the new polymer.

(iv) Give reagents and conditions for step II.

[9]

(b) **Fig.7.2** shows the structure of two organic comppounds, **W** and **Z**,



W **Z**

Fig.7.2

(i) Draw the structure of

1. the compound formed when **W** and **Z** react in aqueous solution.
2. **W** in a neutral aqueous solution.

(ii) Explain the amphoteric behaviour of **W**.

(iii) State, with reasons, which compound, **W** or **Z**, will form enantiomers.

[6]
[Total: 15]

8 (a) The synthesis of 3 - nitrophenylamine is shown in **Fig. 8.1**.

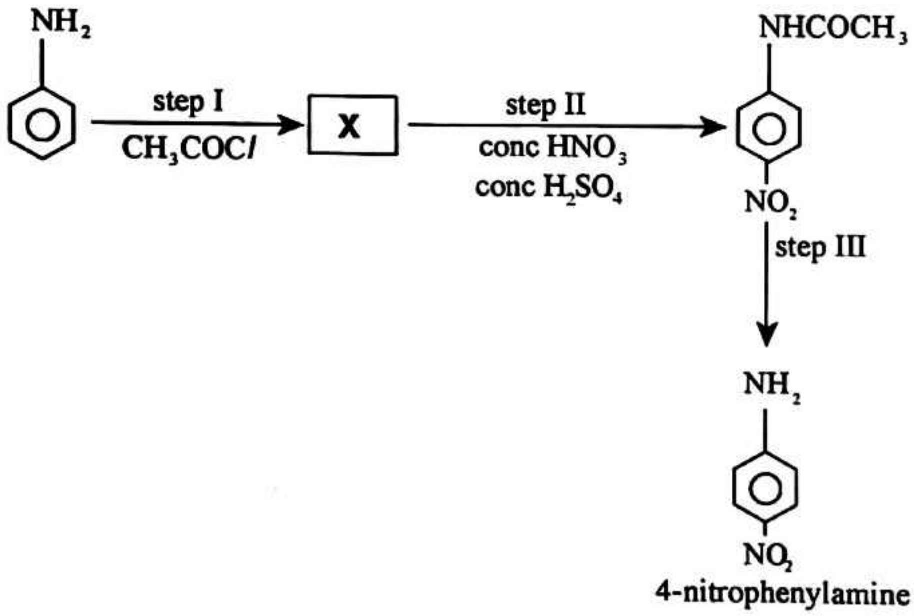


Fig. 8.1

(i) Draw the structure of **X**.

(ii) Name and outline a mechanism for step II.

(iii) Deduce the type of reaction for step III.

(iv) State the reagents and conditions for step III.

[7]

(b) Describe a two step reaction scheme for the formation of phenylamine from benzene. [3]

(c) An organic compound of molecular formula C_4H_8O gave an orange precipitate in the presence of 2,4- DPNH. The compound did not give an observable change when treated with acidified $K_2Cr_2O_7$.

Deduce, with reasons, the

(i) functional group(s) present in the compound,

(ii) possible structural formula of the compound.

[5]
[Total:15]

Section C

Answer any two questions from this section

6 Fig. 6.1 shows how an organic compound can be converted to compounds **G**, **H** and **I**

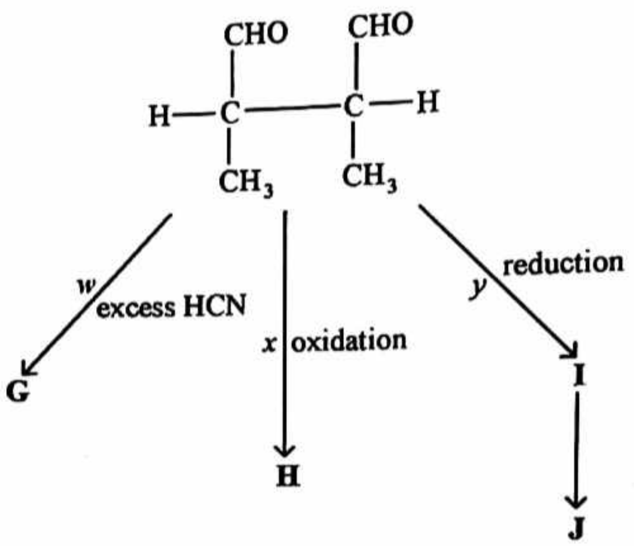


Fig. 6.1

(a) **(i)** State, with reasons, the condition(s) for reaction *w*.

(ii) Name reaction *w*.

(iii) Describe the mechanism for reaction *w*.

[6]

(b) **(i)** State the reagents and conditions for

1. reaction *x*,

2. reaction *y*.

(ii) Draw the structures of **H** and **I**.

(iii) Name compounds **H** and **I**.

(iv) Products **H** and **I** can react to form compound **J**.

Name the type of reaction.

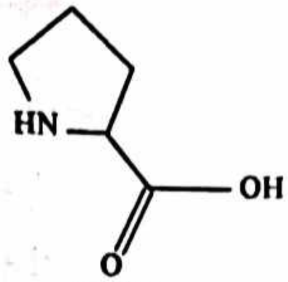
[7]

(c) Describe a one-step laboratory test to distinguish between compounds **H** and **I**.

[2]

[Total 15]

7 (a) Fig. 7.1 shows the structure of an amino acid, proline.



proline

Fig. 7.1

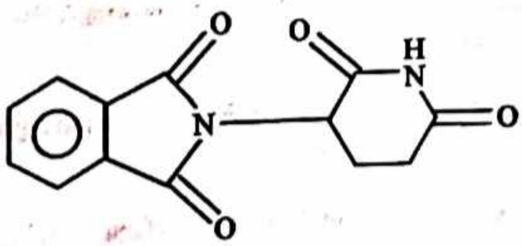
(i) Explain why proline has a high melting point of 223 °C.

(ii) Draw the structure of proline in a neutral solution.

(iii) Write equations to show the buffering action of proline.

[5]

(b) Fig. 7.2 shows the structure of thalidomide, a sedative drug.



thalidomide

Fig. 7.2

(i) Draw the structure of the product formed when thalidomide reacts with

1. $Br_{2(aq)}$,

2. acidic solution.

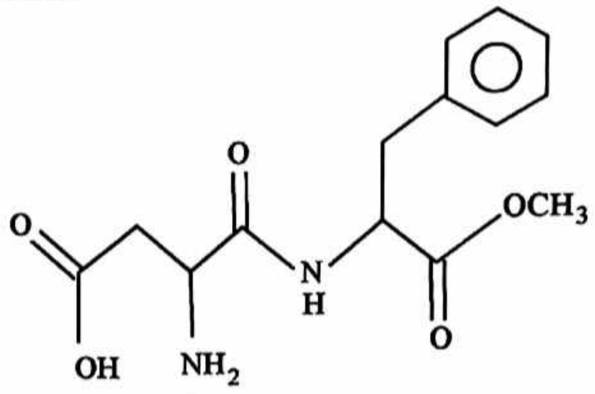
(ii) Explain why thalidomide exists in two different forms.

(iii) Suggest how an unknown sample of thalidomide can be identified using solubility tests.

[1]

[Total: 15]

8 (a) Fig. 8.1 shows the structure of aspartame, a methyl ester of the dipeptide of two amino acids.



aspartame

Fig. 8.1

(i) Write the structures of any **two** functional groups present in aspartame.

(ii) Comment on the solubility of aspartame in water.

(iii) Draw the structural formulae of the products for the complete hydrolysis of aspartame in aqueous HCl.

(iv) Name the **two** amino acids from which aspartame is made.

[8]

(b) Compare the basic strengths of NH_3, $CH_3CH_2NH_2$ and $C_6H_5NH_2$. [3]

(c) The disposal of condensation polymers in landfill sites is more environmentally acceptable than the disposal of addition polymers.

(i) Distinguish between a condensation polymer and an addition polymer.

(ii) Explain why the disposal of condensation polymers in landfill sites is more acceptable.

[4]

[Total: 15]

Section C

Answer any two questions from this section

6 **Fig. 6.1** shows how an organic compound can be converted to compounds **G**, **H** and **I**.

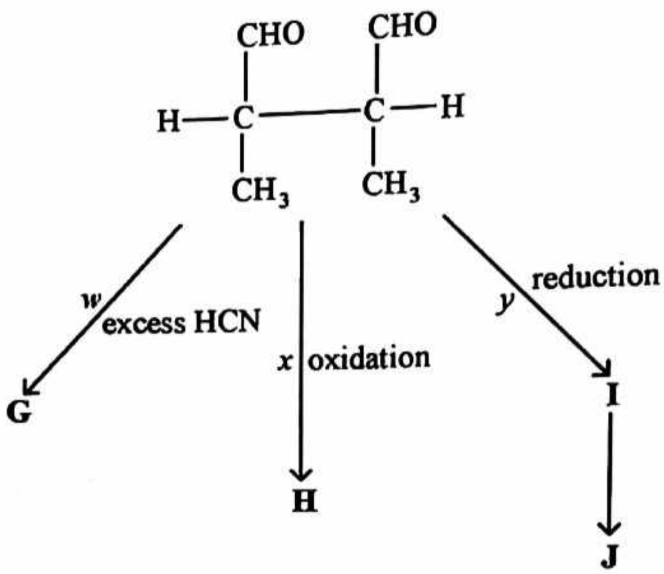


Fig. 6.1

(a) **(i)** State, with reasons, the condition(s) for reaction w.

(ii) Name reaction w.

(iii) Describe the mechanism for reaction w.

[6]

(b) **(i)** State the reagents and conditions for

1. reaction x,
2. reaction y.

(ii) Draw the structures of **H** and **I**.

(iii) Name compounds **H** and **I**.

(iv) Products **H** and **I** can react to form compound **J**.

Name the type of reaction.

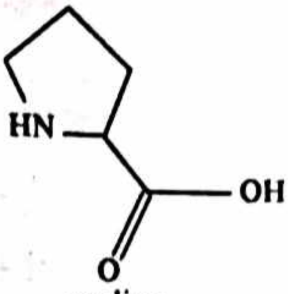
[7]

(c) Describe a one-step laboratory test to distinguish between compounds **H** and **I**.

[2]

[Total 1:

7 (a) Fig. 7.1 shows the structure of an amino acid, proline.



proline

Fig. 7.1

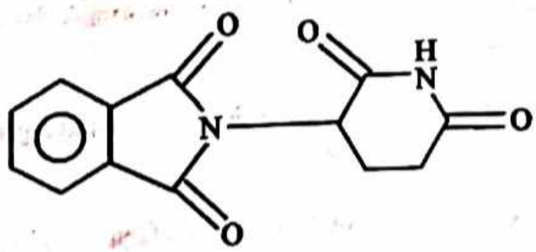
(i) Explain why proline has a high melting point of 223 °C.

(ii) Draw the structure of proline in a neutral solution.

(iii) Write equations to show the buffering action of proline.

[5]

(b) Fig. 7.2 shows the structure of thalidomide, a sedative drug.



thalidomide

Fig. 7.2

(i) Draw the structure of the product formed when thalidomide reacts with

1. $Br_{2(aq)}$.

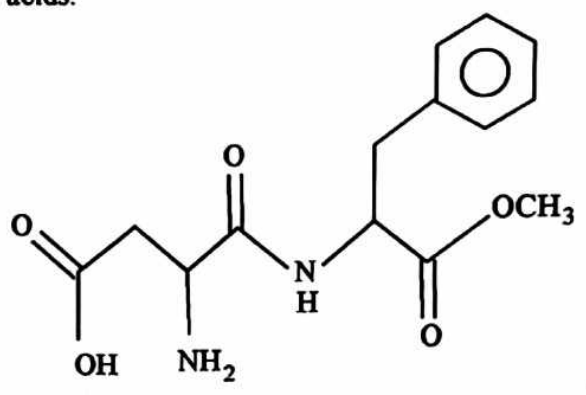
2. acidic solution.

(ii) Explain why thalidomide exists in two different forms.

(iii) Suggest how an unknown sample of thalidomide can be identified using solubility tests.

[1]
[Total: 1?]

8 (a) Fig. 8.1 shows the structure of aspartame, a methyl ester of the dipeptide of two amino acids.



aspartame

Fig. 8.1

(i) Write the structures of any **two** functional groups present in aspartame.

(ii) Comment on the solubility of aspartame in water.

(iii) Draw the structural formulae of the products for the complete hydrolysis of aspartame in aqueous HC*l*.

(iv) Name the **two** amino acids from which aspartame is made.

[8]

(b) Compare the basic strengths of NH_3, $CH_3CH_2NH_2$ and $C_6H_5NH_2$. [3]

(c) The disposal of condensation polymers in landfill sites is more environmentally acceptable than the disposal of addition polymers.

(i) Distinguish between a condensation polymer and an addition polymer.

(ii) Explain why the disposal of condensation polymers in landfill sites is more acceptable.

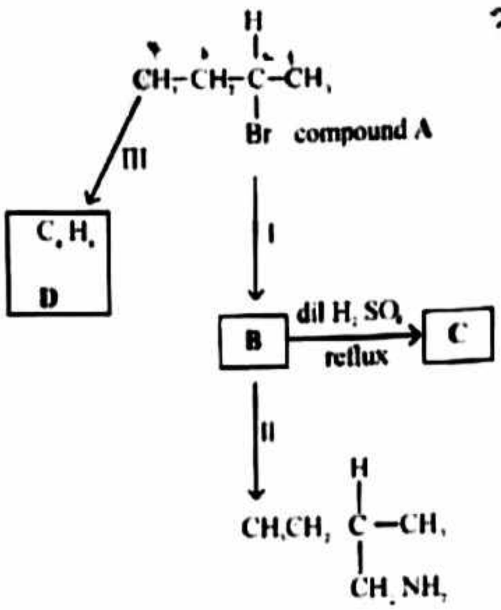
[4]
[Total: 15]

Section C

Answer any two questions from this section.

6 (a) The reaction scheme shown in **Fig. 6.1** shows how compound A can be converted to other organic compounds.

2-bromo butane



CN

NaOH + ethanol

OH

KCN

$CH_3CH_2-C(H)(CN)-CH_3$

Fig. 6.1

(i) Name compound A.

LiAlH4

(ii) Write reagents and conditions for reaction

1. I,

2. II

C–C=C–C

NaOH + ethanol

(iii) Deduce the structural formula of

1. B,

2. C

$CH_3-CH_2-C(CH_3)(H)-Br$

$CH_3-CH_2-C(Br)(CH_3)-Br$

[5]

[Turn over

$CH_3-CH_2-CH_2$

(b) Reaction III forms, three isomeric alkenes of **D**.

(i) State reagents and conditions for reaction **III**.

(ii) Draw the structural formulae of the three isomers of **D**. [5]

(c) Another compound, **E**, which is isomeric with **D** is branched and does not exhibit *cis-trans* isomerism.

(i) Deduce the structural formula of compound **E**.

(ii) Draw the structural formula of the organic products produced when **E** reacts with

1. cold alkaline $KMnO_4$,

2. hot alkaline $KMnO_4$.

(iii) State the observations made in each of the reactions in (ii). [6]
[Total:15]

7 (a) (i) Draw the displayed formula for propylamine

(ii) Comment on the acid-base nature of propylamine.

(iii) Suggest, with reasons, how the pH of propylamine compares with that of 2-aminopropanoic acid. [5]

(b) (i) Define the term *chiral molecule*.

(ii) State the type of isomerism shown by 2-aminopropanoic acid.

(iii) Draw the structures of the isomers. [4]

(c) (i) Explain the term *Zwitterion*.

(ii) Give the formula of the species of 2-amino propanoic acid in aqueous solution at

1. pH 2.

2. pH 12. [3]

(d) Describe and explain how 2-aminopropanoic acid can act as a buffer. [2]
[Total:15]

8 (a) The structural formula of an organic pesticide is shown in Fig. 8.1.

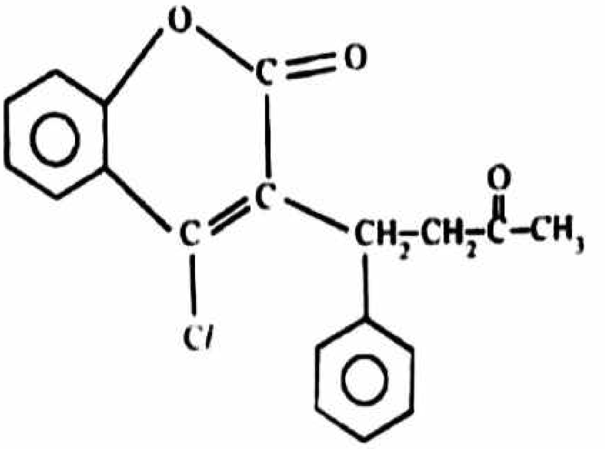


Fig. 8.1

(i) Deduce the molecular formula of the pesticide.

(ii) Identify any **three** functional groups present in the pesticide.

[4]

(b) (i) Draw the structural formula of the organic products formed when the pesticide reacts with

1. Br_2 in an inert solvent,
2. PCl_5.
3. hot NaOH;
4. HCN.

(ii) Name the type of reaction undergone in each of the reactions in (i). [8]

(c) A mass of x g of the pesticide reacted with sodium metal to produce 60.00 cm^3 of hydrogen gas.

Determine the value of x.

[3]
[Total:15]

Section C

*Answer any **two** questions from this section*

6 (a) **Fig. 6.1** shows how methy-2-methylpropenoate can be synthesised.

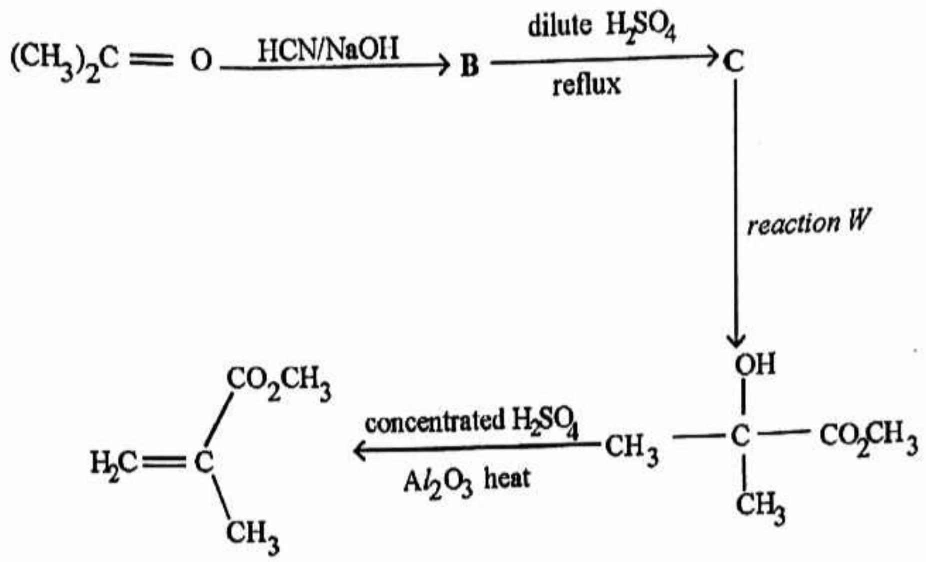


Fig. 6.1

(i) Describe the reaction mechanism for the formation of **B**.

(ii) Draw the structural formula of **C**.

(iii) Name the organic reagent of *reaction W*. [5]

(b) Methyl-2-methylpropenoate polymerises to form a transparent thermoplastic glass.

Write a chemical equation for the polymerization process. [2]

[Turn over

(c) **Fig. 6.2** shows the structure of Taxol.

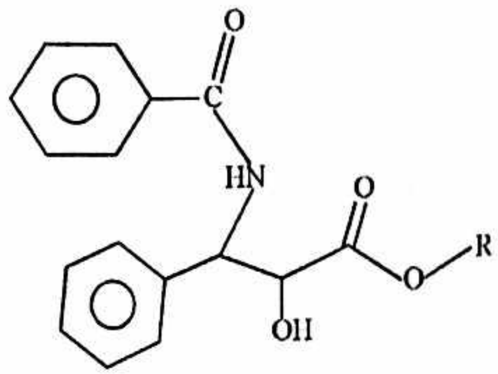


Fig. 6.2

(i) Name any **two** functional groups in Taxol.

(ii) State, with reasons, the type of isomerism that exists in Taxol.

(iii) Draw the structural formulae of the organic products formed when Taxol is treated with hot aqueous sulphuric acid.

[6]
[Total:15]

7 (a) The displayed formula of acrolein, is shown in **Fig. 7.1**.

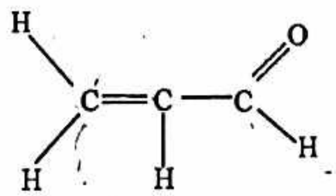


Fig. 7.1

(i) Describe any **two** chemical tests that can be carried out to show the presence of acrolein.

(ii) Draw two repeat units of a polymer produced when acrolein polymerises.

(iii) Explain why acrolein undergoes the type of polymerisation described in (ii).

[6]

(b) The reaction scheme in **Fig. 7.2** shows four reactions of acrolein to produce compounds W, X, Y and Z.

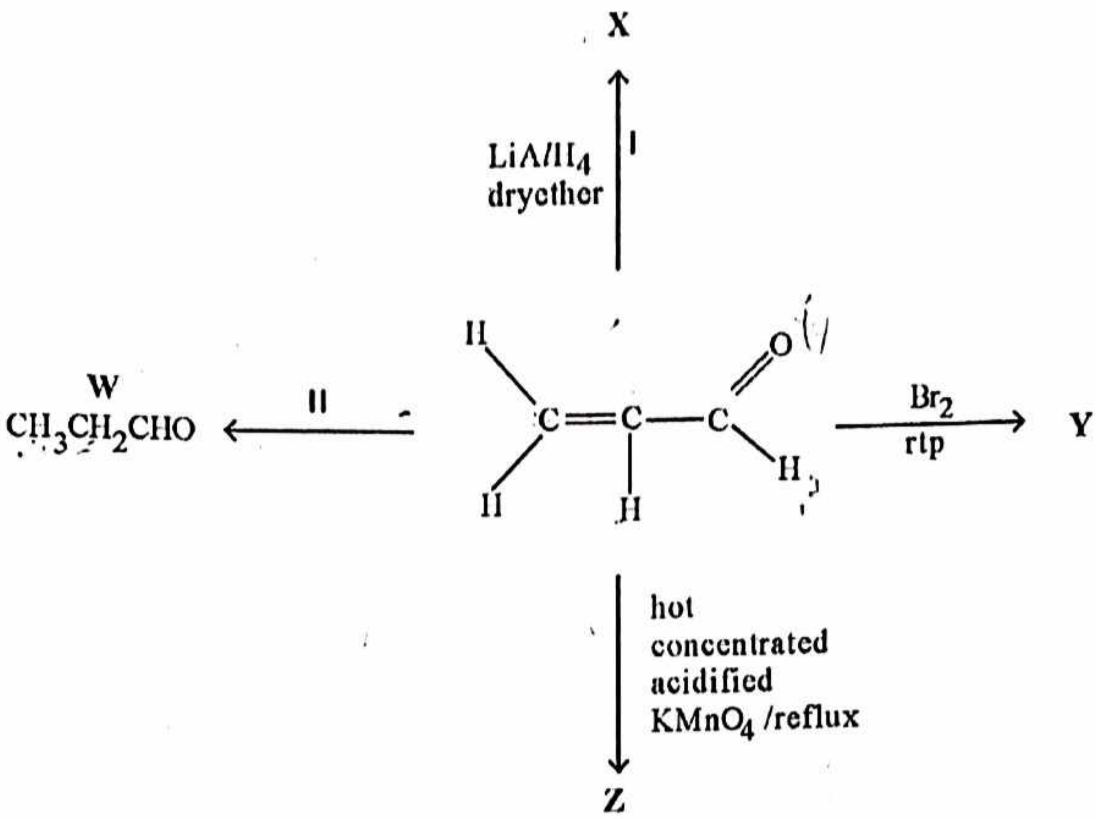


Fig. 7.2

(i) Name the type of reaction taking place in step **I**.

(ii) Draw the displayed formula of

1. **X**,

2. **Z.**

(c) Compound **W** reacts with HCN in the presence of a base.

(i) Name the type of reaction. [1]

(ii) Explain why the rate of reaction is catalysed by the presence of a base.

(iii) Construct a mechanism for the reaction.

[6]
[Total:15]

[Turn over

Section C

Answer any two questions from this section

6 **(a)** Carbon-carbon double bonds in benzene are not typical carbon-carbon double bonds.

(i) Describe and explain any two chemical reactions that can be used to prove this observation.

(ii) Explain why benzene is generally inert.

[4]

(b) **Fig. 6.1** shows the structure of estradiol.

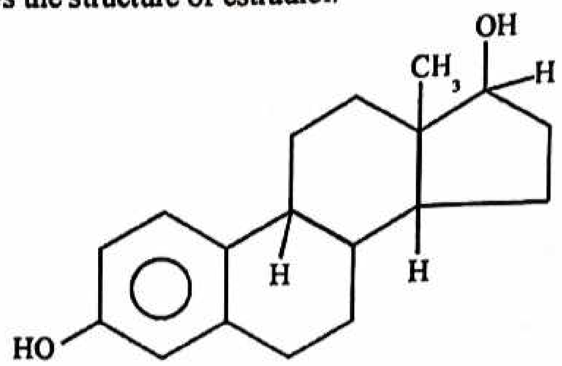


Fig. 6.1

(i) State

1. the number of chiral centres in estradiol,

2. with reasons, the acid base nature of estradiol.

(ii) Explain why estradiol

1. readily dissolves in water,

2. has a high boiling point.

(iii) Draw the structure of the organic product formed when estradiol reacts with

1. ethanoyl chloride,

2. aqueous bromine in the dark.

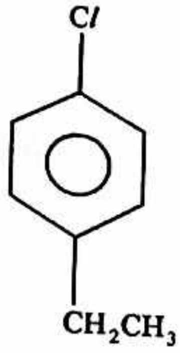
(iv) State the observations made in each of the reactions in **(iii)**.

[9]

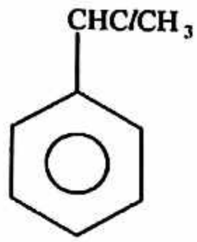
(c) Suggest methods that can be used to reduce the destruction of the ozone layer through pollution by chlorine containing compounds. [2]

[Total: 15]

7 **(a)** **Fig. 7.1** shows the structures of two organic compounds **A** and **B**.



A



B

Fig. 7.1

(i) State the type of isomerism shown by **A** and **B**.

(ii) Compare the reactions of **A** and **B** with $NaOH_{(aq)}$.

[4]

(b) The emperical formula of an organic compound **R** is CH_2O. Chemical tests carried out on **R** and the observations made are shown in **Table 7.1**.

Table 7.1

test	observation
R was warmed with alkaline iodine	yellow precipitate formed
2,4-DNPH was added to **R**	no observable changes
Na_2CO_3 was added to **R**.	bubbles of gas produced

(i) Deduce the functional groups in **R**.

(ii) Further chemical tests carried out on **R** showed that **R** is a chiral molecule and has a molecular mass of 90.08.

Deduce the structural formula of **R**.

[7]

[Turn over

(c) **Fig. 7.2** shows some of the reactions of 1,4 –dichlorobutane.

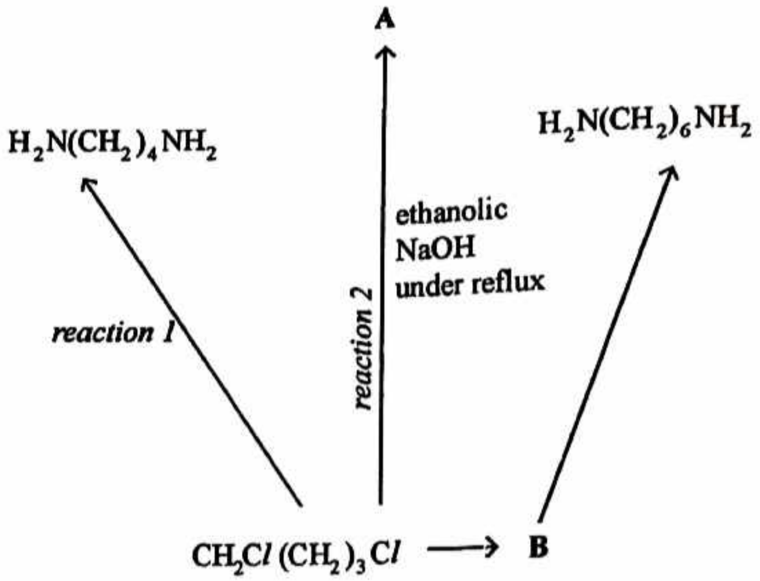


Fig. 7.2

(i) Name

1. *reaction 1*,

2. *reaction 2*.

(ii) Draw the structural formula of **A**.

(iii) Deduce the conditions needed for the formation of **B**.

[4]
[Total: 15]

8 An organic compound, with structural formula $CH_3COCH_2CH_2COOCH_3$ undergoes the reaction scheme shown in **Fig. 8.1**.

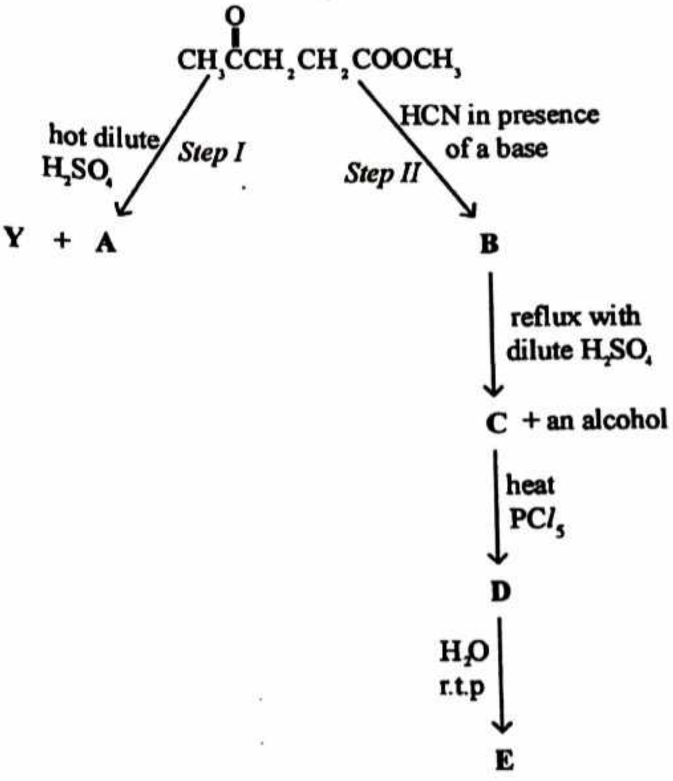


Fig. 8.1

(a) **(i)** Draw the structural formula for **A**, **Y** and **B**.

(ii) Name the type of reaction undergone in

1. step I,

2. step II.

[5]

(b) Write the reaction mechanism for *step II*. [3]

(c) Suggest the structural formula for

(i) **C**,

(ii) **D**,

(iii) **E**.

[3]

(d) (i) State, with reasons, the acid/base nature of **E**.

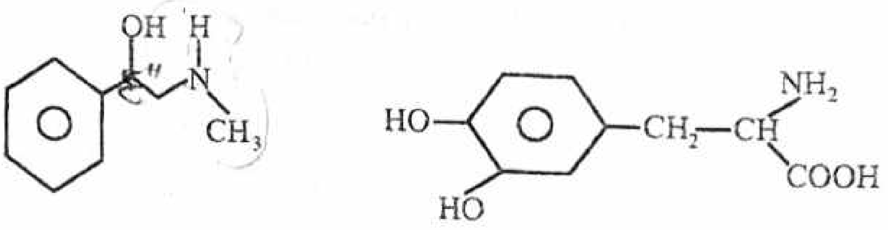
(ii) Comment on the use of solution **E** as a component of tear gas used by law enforcement agencies.

[4]
[Total:15]

Section C

Answer any two questions from this section

6 (a) The structures of two organic compounds **Y** and **Z** are shown in **Fig. 6.1**.



Y Z

Fig. 6.1

(i) Draw the isomers of **Y** and state the type of isomerism shown.

(ii) Draw the displayed structural formula of the organic product formed when

1. **Y** is warmed with acidified $KMnO_{4(aq)}$,

2. sodium metal is added to **Z**.

3. **Z** is reacted with PCl_5.

(iii) Compound **Z** is a drug used in the treatment of human diseases.

Comment on the

1. acid-base nature of **Z**,

2. long term use of the drug on the human body.

[9]

(b) **Fig. 6.2** shows the structure of glyoxal, an organic compound used as a disinfecting agent.

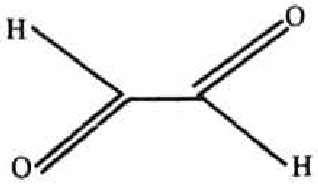


Fig. 6.2

(i) Explain why glyoxal is suitable for its use.

(ii) Describe and explain how glyoxal can be prepared from ethene in a two step reaction.

(iii) Explain why glyoxal exists as a dimmer at high concentration.

[6]

[Total:15]

Fig. 7.1 shows the structural formula of cholesterol.

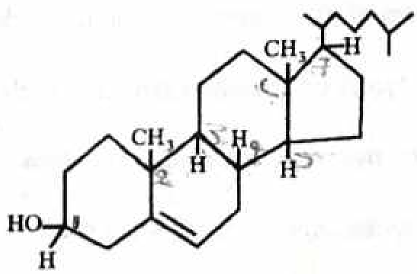


Fig. 7.1

(a) **(i)** Name the functional groups in cholesterol.

(ii) Write the molecular formula of cholesterol.

(iii) State the number of chiral centers in cholesterol.

(iv) State and explain how the reaction between cholesterol and sodium metal converts it into a more soluble compound.

[6]

(b) Describe and explain the changes that occur when cholesterol reacts with

(i) Br_2 in the dark,

(ii) hot acidified potassium permanganate.

[7]

(c) Name the type of reaction that occurs in each of the reactions in **(b)**. [2]

[Total: 15]

P4

[Turn over

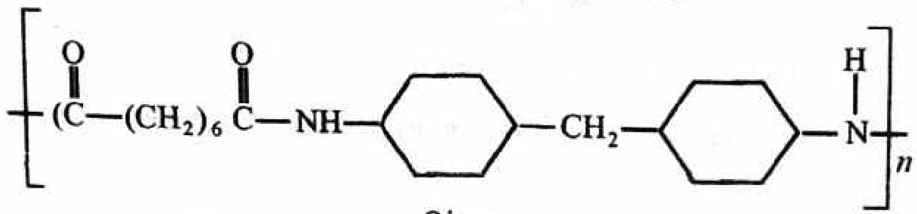
(a) Describe the differences between addition and condensation polymers in terms of

(i) hydrophobic/hydrophilic nature,

(ii) difficulty of reactivity.

[2]

(b) **Fig. 8.1** shows the structure of a synthetic polymer, Qiana.



Qiana

Fig. 8.1

(i) Identify the type of polymerisation used to produce Qiana.

(ii) Deduce the structures of the monomers used to produce the polymer.

(iii) Explain why the polymer can produce strong and elastic material.

(iv) Suggest a use for the material produced from Qiana.

(v) Qiana deteriorates when exposed to alkaline conditions for long periods of time.

1. Explain why the material deteriorates.
2. Write an equation to illustrate the process that leads to its deterioration.

[9]

(c) Spiders produce a polymer, spider silk, which has similar bonds as Qiana.

(i) State **one** similarity and **one** difference between spider silk and Qiana.

(ii) Suggest **two** ways in which spider silk is important in the life of a spider.

[4]
[Total:15]

Section C

*Answer any **two** questions from this section.*

6 (a) A 1.00 g sample of hydrocarbon, **Q**, was vapourised at 77 °C and 100 kPa. The gas produced occupied a volume of 220 cm^3.

(i) Calculate the relative molecular mass of **Q**.

(ii) Deduce the molecular formula of **Q**, given that its empirical formula is C_5H_6. [5]

(b) When **Q** was refluxed with acidified concentrated potassium manganate (VII), a mixture of two carboxylic acids was obtained. One of the acids has the structural formula shown in **Fig. 6.1**.

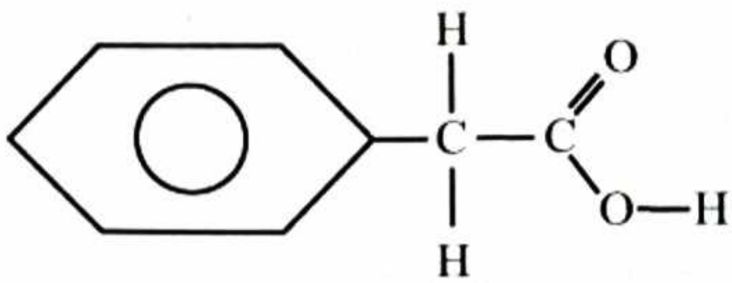


Fig. 6.1

(i) Describe the observation made when the concentrated acidified $KMnO_4$ is added to **Q**.

(ii) State the type of reaction that occurs between **Q** and acidified $KMnO_4$.

(iii) Suggest the formula of the other acid.

(iv) Deduce the structural formula of **Q**.

(v) Draw the structural formula of a hydrocarbon which is isomeric with **Q**. [7]

(c) (i) Define the term *polymerisation*.

(ii) Name the type of polymerisation that **Q** undergoes.

(iii) Draw a segment of a polymer made from **Q**.

[3]

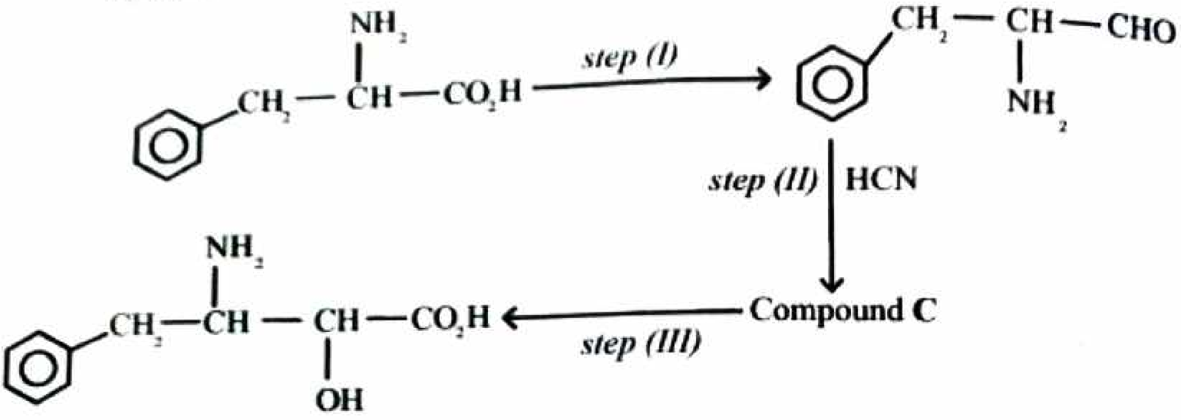
[Total:15]

CaCO3 + NH4 SO4

NH3 + CaSO4 + CO2

NH3 + O2 →

7 **(a)** The reaction scheme shown in **Fig. 7.1** shows how an organic compound, **B**, can be synthesized.



Compound **B**

Fig. 7.1

(i) State the number of chiral centres in **B**.

(ii) Suggest reagents and conditions for

1 *Step I*,

2 *Step III*.

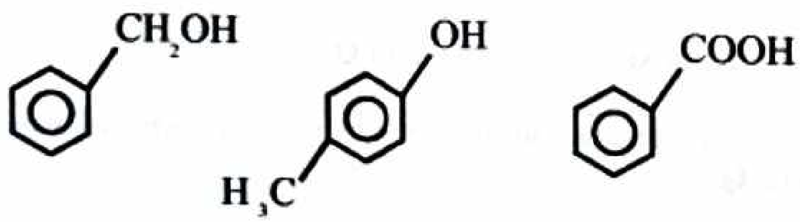
(iii) Name the type of reaction in

1 *Step I*,

2 *Step III*.

(iv) Give the structure of **C**. [6]

(b) **Fig. 7.2**. shows the structural formulae of three organic compounds **P**, **Q** and **R**.



P **Q** **R**

Fig. 7.2

(i) Explain the relative acidities of **P**, **Q** and **R**.

(ii) Describe a chemical test that can be used to distinguish **R** from **P** and **Q**. [6]

(c) The structure of an organic compound **M**, is shown in **Figure 7.3**.

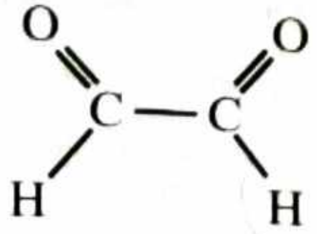


Fig. 7.3

(i) Give the structure of the organic product formed when **M** reacts with

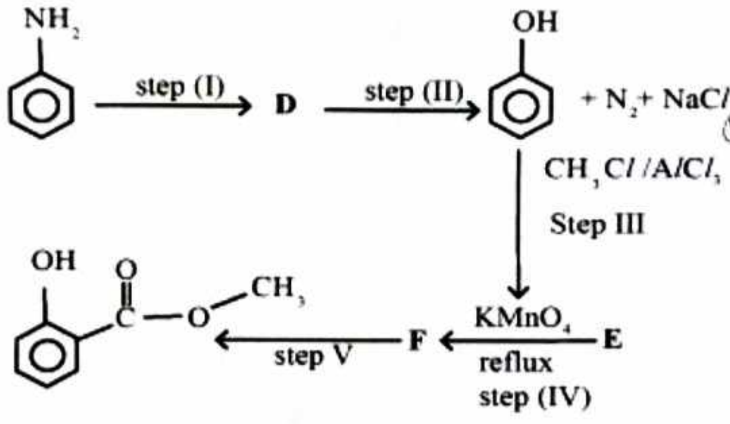
1 $LiAlH_4$,

2 2,4 - dinitrophenylhydrazine.

(ii) Name the type of reaction undergone by **M** when it reacts with 2,4 - dinitrophenylhydrazine. [3]

[Total:15]

8 **Fig. 8.1** shows how an organic compound, **G** can be synthesised.



Compound **G**

Fig. 8.1

(a) Name compound **G**. [1]

(b) Compounds **Y** and **Z** are isomers of **G** and they react with PCl_5. **Y** reacts with PCl_5 in the ratio 1:1 and **Z** in the ratio 1:2.

(i) Deduce the structural formulae of **Y** and **Z**.

(ii) Name the type of structural isomerism shown.

SECTION C

Answer any two questions from this section

6 (a) Write down a balanced chemical equation for the fermentation of glucose. [2]

(b) Compare the relative acidities of ethanol, phenol and water. [3]

(c) An organic compound **A**(Mr 72) was found to contain 66.7% carbon, 11.1% hydrogen and the rest oxygen only. **A** decolourised bromine water and gave a positive tri-iodomethane test with alkaline aqueous iodine, but did not react with 2.4 DNPH. When catalytically hydrogenated, one mole of **A** absorbed one mole of hydrogen to give **B**. **B** was oxidised by acidified potassium dichromate (VI) to give **C**(Mr 72). **C** did not react with Tollen's reagent.

(i) Determine the molecular formula of A.

(ii) Draw the structural formula of **A**, **B** and **C**.

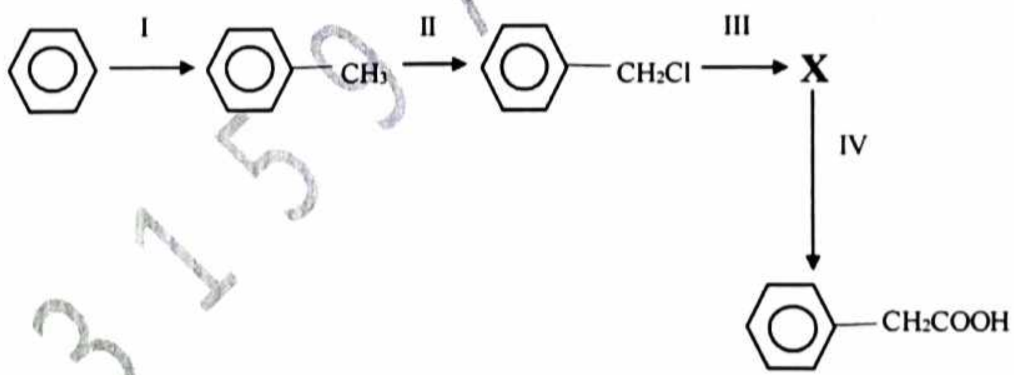
(iii) Name any one functional group present in A and put an asterisk (*) on any chiral centre.

(iv) Name compound A and draw its optical isomers.

[10]

[TOTAL: 15]

7 (a) The reaction scheme below shows how compound Z can be obtained from benzene.



Z

(I) State the reagents and conditions for steps:

1. I
2. II

(II) Explain the mechanism for step I.

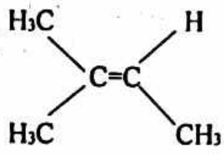
[7]

(b) (i) Suggest the structure of the intermediate X.

(ii) Name reaction IV.

[2]

(c) The structure of compound **P** is shown below



P

(i) Would you expect **P** to show cis-trans isomerism? Explain your answer.

(ii) The reaction of compound **P** with $HBr_{(g)}$ produced one major organic product and traces of another. Draw the structures of these two organic products.

(iii) Suggest the structure of the major product from the two you listed above. Explain your reasoning.

[6]

[TOTAL: 15]

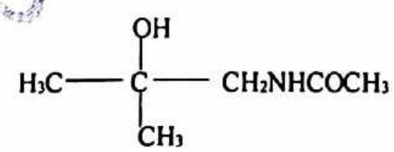
8 (a) Carbonyl compounds react with hydrogen cyanide to form cyanohydrins.

(i) Explain clearly why this reaction should be carried out in the presence of a base.

(ii) Explain the mechanism for the reaction above in (a) (i) using any carbonyl compound of your choice.

[5]

(b) Starting form a **suitable** carbonyl compound of your choice, and using a cyanohydrin as an intermediate, device a 3-stage synthesis of compound **A**.



A

Suggest the reagents and conditions for each step, and draw the structural formula of every intermediate compound.

[6]

(c) Using the general structure of an amino acid, $H_2NCH(R)COOH$, explain the amphoteric behaviour of amino acids.

[2]

(d) Which one is a stronger acid, ethanoyl chloride or chloro-ethanoic acid. Explain. [3]

[TOTAL: 15]

Section C

*Answer any **two** questions from this section.*

6 (a) (i) Explain the term *substitution reaction*.

(ii) State and explain the changes in the volatility of alkanes as the number of carbon atoms increase.

(iii) Distinguish between addition polymerisation and condensation polymerisation. [5]

(b) **Fig 6.1** shows how organic compounds M and N can be synthesized.

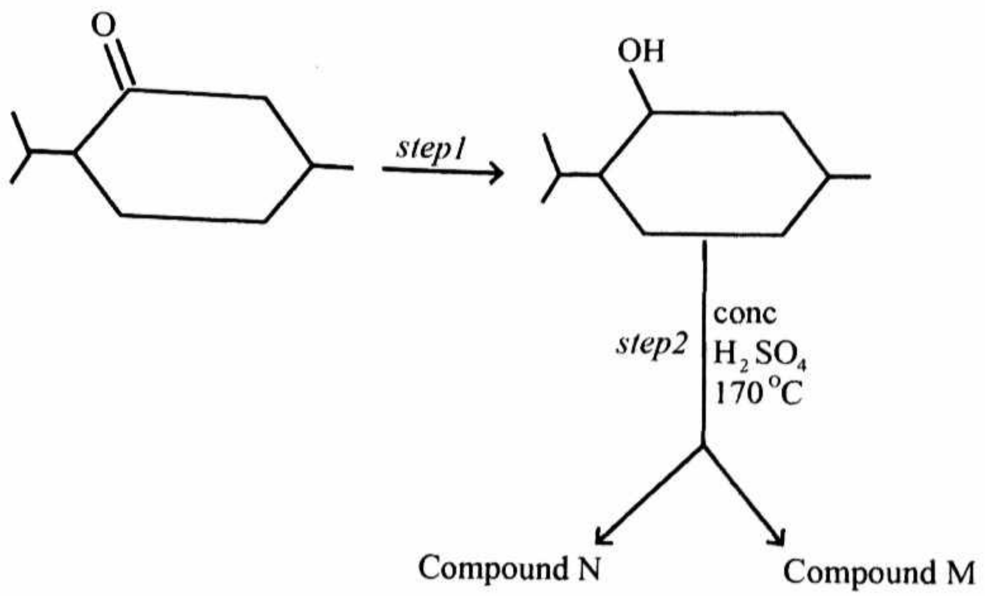


Fig. 6.1

1. Give the structures of M and N.

2. Suggest the reagents and conditions for *step 1*. [4]

(c) (i) Give the mechanism for the chlorination of benzene.

(ii) Outline the differences in the reaction between Cl_2 and ethene and the chlorination of benzene. [6]

[Total:15]

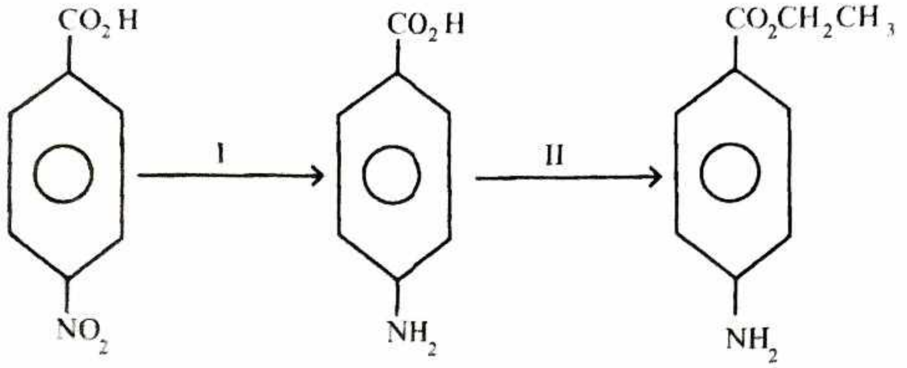
7 **(a)** **(i)** Define the term *cracking*.

(ii) Write an equation to show cracking of $C_{16}H_{34}$.

(iii) State the conditions for cracking.

(iv) Give one advantage of cracking. [5]

(b) **Fig. 7.1** shows how benzocaine can be synthesized from 4-nitrobenzoic acid.



benzocaine

Fig. 7.1

(i) Suggest the reagents and conditions for steps I and II.

(ii) Give the organic product formed when benzocaine react with

1. dil H_2SO_4 under reflux,
2. Br_2.

(iii) Compare the solubility of Benzocaine and that of 4-nitrobenzoic acid. [6]

(c) **Fig. 7.2** shows the structure of an organic compound Z.

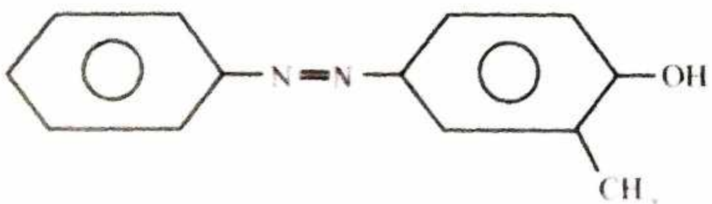


Fig. 7.2

(i) Draw structures of compounds used to prepare Z.

(ii) Name the type of reaction used in preparing Z.

(iii) Comment on the solubility of Z. [4]

[Total: 15]

8 (a) Mandelic acid can be synthesized from benzaldehyde in a two-step reaction as shown in **Fig. 8.1**.

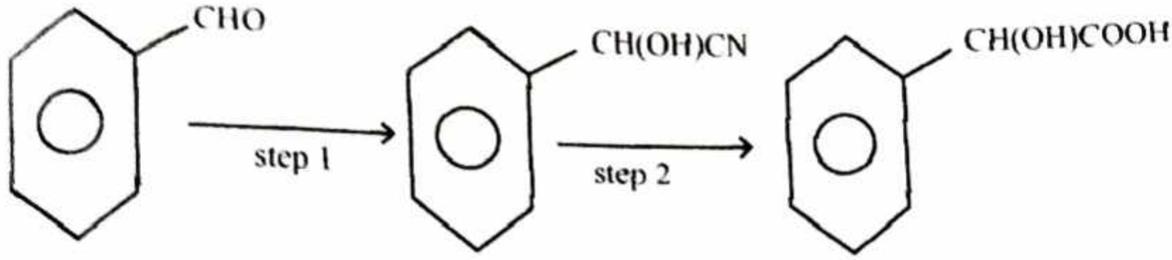


Fig. 8.1

(i) State the type of isomerism exhibited by Mandelic acid.

(ii) Draw structures of the isomers of Mandelic acid.

(iii) Name the type of reaction occuring in step 2.

(iv) Give the reagents and conditions required for

1. step 1,
2. step 2.

(v) State the significance of the reaction occuring in step 1 in organic synthesis. [9]

(b) Describe two test tube reactions that can be used to distinguish between benzaldehyde and Mandelic acid. [4]

(c) **(i)** State the reducing agent that can be used to reduce the intermediate Mandelonitrile to another compound D.

(ii) Draw the structure of compound D. [2]

[Total:15]